

Thermal kinetic, mechanical and wear properties of glass fiber reinforced polyamide 6 thermoplastic composites

Kamlendra Vikram¹ | Sumit Pramanik¹  | Shubrajit Bhaumik²

¹Functional and Biomaterials Engineering Lab, Department of Mechanical Engineering, College of Engineering and Technology, SRM Institute of Science and Technology, Kattankulathur, Chennai, Tamil Nadu, India

²Tribology and Interactive Surface Research Laboratory (TRISUL), Department of Mechanical Engineering, Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Chennai, Tamil Nadu, India

Correspondence

Sumit Pramanik, Functional and Biomaterials Engineering Lab, Department of Mechanical Engineering, College of Engineering and Technology, SRM Institute of Science and Technology, Kattankulathur, Chennai 603203, Tamil Nadu, India.
Email: sumitpr@srmist.edu.in; prsumit@gmail.com

Shubrajit Bhaumik, Tribology and Interactive Surface Research Laboratory (TRISUL), Department of Mechanical Engineering, Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Chennai 601103, Tamil Nadu, India.
Email: shubrajit.projects@gmail.com

Abstract

Polyamide 6 (PA6) thermoplastic-based composites are often used in automobile, aerospace, marine, and biomedical applications. Here, PA6 and its composites with varied chopped glass fiber (GF) concentrations of 5, 10, and 20 mass% were fabricated by injection molding. The specimens were characterized by structural, thermal kinetic, mechanical, as well as tribological analyses. Tribological performance was evaluated in dry pin-on-disc test apparatus with three different levels of time, speed, and load factors. The PA6/GF composites showed high decomposition activation energy compared to pure PA6. Thermal activation, E_a increased to 46.16% with increasing of GF content up to 10 mass % in the composites compared to PA6. PA6/5GF exhibited best mechanical properties (e.g., strength = 44.46 MPa and Hardness = 45.2). Addition of GF in PA6 significantly improved tribological properties, coefficient of friction (COF) and specific wear rate (WR) besides thermal stability and mechanical properties of the composites. Especially, COF and WR of PA6/5GF composite decreased by 17%–65% and 68%–92%, respectively, compared to PA6. ANOVA statistical results determined the significant contribution of the tribological factors to the performance metrics of the composites for COF and WR. Therefore, the PA6/5GF composite would highly be useful for tie-rod gears and bearings as a crucial structural component in automotive bodywork.

KEYWORDS

friction, kinetics, mechanical properties, polyamides, thermogravimetric analysis (TGA), wear and lubrication

1 | INTRODUCTION

The study on application of friction, lubrication, and wear is part of a branch in engineering science known as tribology, which focuses on interacting surfaces while in motion. Due to its interdisciplinary applications, research on tribology is explicitly explored by many multiple sub-disciplines from physics, chemistry, mathematics, and materials science to engineering.¹ In this context, plastic materials have tremendously been used in the industrial field in past decade because of their typical tribological

and mechanical properties over metallic materials.² The main advantage of plastic material is low weight, low price for high-volume production, reduce noise level, and good self-lubricating properties.³ However, plastic materials has also shown some specific disadvantages compared to metallic materials such as, inferior mechanical properties, less thermal conductivity, and lower operating temperature.^{4,5} The polymer matrix composite (PMC) which includes composites based on polyamide 6 (PA6) thermoplastic has undergone a significant amount of study in the past few years.^{6,7} Due to its exceptional

physical and mechanical capabilities, PA6 has become an important engineering plastic as it has widely been utilized in a variety of engineering products, including automobile parts, electrical equipment, and packaging materials. However, PA6 has a number of demerits, including poor dimensional stability, low thermal deformation temperature, high susceptibility to notch impact, and low thermal deformation temperature. Hence, the mechanical qualities of the engineering plastics must also be enhanced. Many studies have been conducted on the tribological behavior of other engineering plastics including, polyamide (PA) and high-density polyethylene (HDPE) and their composites.^{8–11} Depending on the application, glass fiber (GF) has widely been used as reinforcement in order to improve the mechanical and tribological properties of the polymers and to reduce the cost when compare to other materials of similar strength.^{12,13} The GF is the most popular reinforcement material used in the various field such as automotive, marine, and aerospace, and so on owing to its high tensile strength, outstanding electrical insulation, good chemical resistance, great durability, excellent heat resistant, great fire-resistant, good thermal conductivity, and high economical and dimensional stability. Some commonly used GFs are E-GF (in electrical applications), C-GF (in corrosive environments), S-GF (in structural applications). Due to the current financial crisis, it has also been attempted to make hybrid polymer matrix composites (HPMCs) using some natural fibers, including cotton, leaf, jute, banana, sisal, bamboo, nanoclay, graphene oxide, and kenaf.¹⁴ However usually, HPMC are not economically cheap. Several studies have been conducted to better understand the influence of GF component, content, and orientation, as well as the fabrication method and test conditions on the tribological properties. Grozdanov and Bogoeva studied on hybrid yarns based carbon fiber (CF)/PA6 composites and they showed that the CF can significantly enhance the mechanical properties of PA6 composites.¹⁵ In another study, long glass fiber (LGF)/PA6 composite was developed by reinforcing LGF into impregnated PA6 oligomer melt followed by solid-state polymerization.¹⁶ The mechanical properties of the LGF/PA6 composites were noticeably enhanced. The processing parameters also have some influence on the mechanical and microstructural properties of nylon/glass composites.¹⁷ The combination of PA6/clay and GF/PA6/clay nanocomposites with various reinforcements produced synergy effect.¹² Further study on PA6/GF composites revealed that GF can significantly decrease the specific wear rate (WR) and coefficient of friction (COF).^{18–20} Zaghoul et al. conducted a research on the tribological performance of GF reinforced with PA6 in highly abrasive conditions. They found that GF had best

ability to minimize the COF. At low photovoltaic (PV) settings, the WR of PA6/25GF decreased due to the inclusion of GF, indicating its excellent resistance to wear.²¹ Uysal et al. compared the mechanical and tribological characteristics of composites composed of polyketone reinforced with GF, and a combination of polyketone (PK) and PA6. They found that the COF and WR of PK/PA6/GF30 composites were enhanced besides impact resistances by increasing the PK segment.²² Khakbaz et al. conducted a study to analyze the tribological and mechanical properties of GF-reinforced PA6 composites after being subjected to hydrothermal aging. But, they obtained that the inclusion of up to 25 wt.% GF in PA6-based composites could lead to a noticeable increase in abrasion wear mass loss and a decrease in COF while the tensile modulus and strength of unfilled PA6 and PA6-GF composites were decreased by hydrothermal aging of the specimens.²³ Zaghoul et al. examined the impact of counter-face grit size and lubricant on the abrasive wear characteristics of thermoplastic polymers reinforced with GF. They found that the addition of molybdenum disulfide (MoS_2) resulted in a notable improvement in the materials' resistance to wear across all grit sizes and their 33GF PA6 material exhibited the highest level of wear resistance, surpassing both 25GF and 50GF.²⁴ A surface treatment of CF can also effectively increase the adhesion between the interfaces of CF and PA6 matrix composites.²⁵ The tribological characteristics of PA6 reinforced with aramid fiber and nano- MoS_2 particles exhibited the multilayer laminar film of worn debris on the wear scar, the MoS_2 particles transported prior to PA6, and decreased in both COF and WR following the addition of nano- MoS_2 .²⁶ However, no study was reported on the correlation of structural, mechanical thermal, and tribological properties on PA6 composites till date.

Therefore, the main purpose of this article is to determine the effect of 0.1–0.5 mm chopped GF (5, 10, and 20 mass%) in PA6 composites on the structural, mechanical, thermal, and tribological behaviors. In dry condition, the wear behavior of a pin-on-disc was extensively explored with optimization. However, the microstructural and thermal kinetics of PA6 matrix, which are mainly influenced by the processing conditions, determine the mechanical properties of fiber reinforced PA6 composites and they have not been investigated precisely till date. Hence, this study focused to determine a correlation among the mechanical properties, fracture surfaces, hydrophilicity, and thermal stability with the tribological properties of these composites. Hence, the present study aimed to select the appropriate filler for a PA6 matrix that can withstand high load, speed, and time.

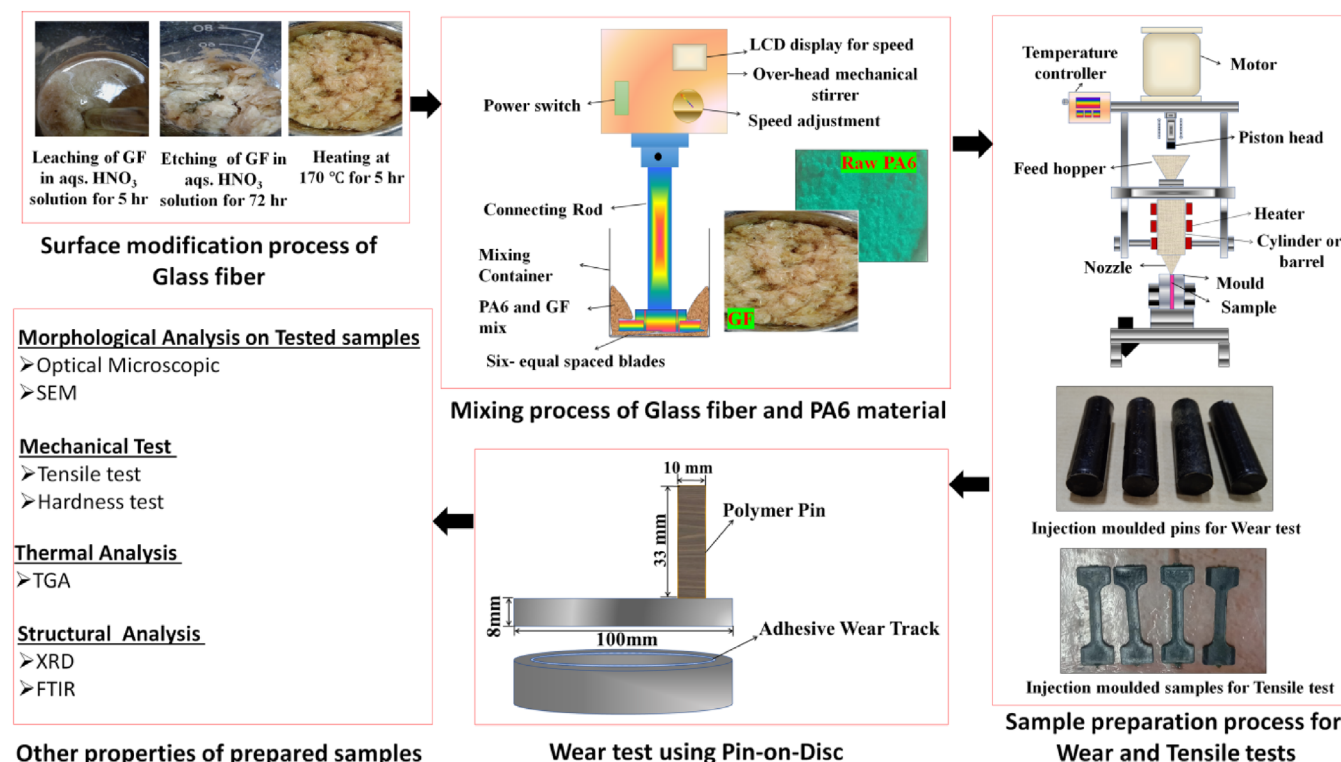


FIGURE 1 Schematic flow diagram of PA6 and PA6/GF (5, 10, and 20 mass %) composite sample preparation for wear and tensile tests. GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

2 | MATERIALS AND METHODS

2.1 | Material details

In this investigation, the most important ingredients used are commercial PA6 powder and E-glass fiber (GF). The rotating disc ($\phi 100 \text{ mm} \times 10 \text{ mm}$) was prepared with EN31 steel material of hardness 35 HRC. The melting temperature (T_m) of commercial PA6 polymer powder was determined by differential scanning calorimetry (DSC) as 221°C found in our previous study.²⁷

A pure commercial PA6 powder with the molecular weight (MW), density (ρ), and melting temperature (T_m) of 20,000 g/mol, 1.14 g/cm^3 , and 221°C , respectively, purchased from Jay Mataji Dye Chem, Ahmadabad, India was used for this present study. The matrix material was the pure PA6 powder and the reinforcement material was etched GF at different concentrations of 5, 10, and 20 mass%. Here, the GFs of diameter in the range of $8\text{--}10 \mu\text{m}$, $\rho \sim 2.54 \text{ g/cm}^3$, and soften temperature $\sim 1200^\circ\text{C}$ were cut into the length of $0.1\text{--}0.5 \text{ mm}$ to be used in the composites after etching.

2.2 | Surface modification of GF

A chemical agent of 3M nitric acid (HNO₃) aqueous solution was used for surface treatments of GF by chemical

etching process to increase the bond strength, surface roughness, and wettability between the GF and PA6.^{28,29}

In this process, the chopped GFs were washed with distilled water followed by chemically leaching in the aqueous HNO₃ solution at room temperature for 5 h, then etching in the aqueous HNO₃ solution at room temperature for 72 h, and further heating at 170°C for 5 h to evaporate the aqueous solution. Finally, the treated GFs were dried at room temperature for 24 h. Figure 1 shows the etching process of E-GF. It was done because it has been found by other study that surface-modified silicon carbide (SiC) nanoparticles contributed a noticeable improvement in the wear properties compared to the untreated SiC nanoparticles reinforced composites.³⁰ The dried etched GFs were used to make the composites for the entire study.

2.3 | Composite preparation

An injection molding machine was used to manufacture a cylindrical ($\phi 10 \text{ mm} \times 33 \text{ mm}$) specimen to evaluate the tribological, mechanical, physical, and thermal properties of PA6 and PA6/GF(5, 10, and 20 mass %) composites. A flow diagram of manufacturing the composites is illustrated in Figure 1 and the composition details of the composites sample are given in Table 1. The etched GF was mixed in raw PA6 powder with the help of stirrer

TABLE 1 Composition details of the composites sample.

Sample code	PA6 (mass%)	GF (mass%)
PA6 raw powder	100	0
E-glass fiber (GF)	0	100
PA6	100	0
PA6/GF5	95	5
PA6/GF10	90	10
PA6/GF20	80	20

Abbreviation: GF, glass fiber.

machine for 2 h of time at a speed of 1000 rpm followed by injection molding. During injection molding, top mold, and bottom mold temperatures were kept at 260 and 230°C, respectively, while the injection pressure was held at 70 MPa and the cooling time was kept constant for all the samples.

2.4 | Experimental details

2.4.1 | Fourier transformation infrared study

Fourier transformation infrared (FTIR) was used to evaluate the relationships between the structure and properties of PA6 and PA6/GF(5, 10, and 20 mass%) reinforcement composites using an IRTracer-100-Shimadzu spectroscope with a resolution of 2 cm⁻¹ over a range of wave number from 500 to 4000 cm⁻¹.

2.4.2 | X-ray diffraction study

Using x-ray diffractometer (XRD) (Xpert powder XRD system, Panalytical), the XRD patterns of the samples were obtained at a scan rate of 2°/min and the scanning angle 2θ varied from 5° to 60°. The crystal structure was investigated using the XRD technique.

2.4.3 | Thermogravimetric analysis

Thermogravimetric analysis (TGA) was carried out using a thermal analysis system (STA7200, Hitachi) to investigate the degradation kinetics and thermal stability of the raw PA6 powder, raw GF, PA6, and PA6/GF(5, 10, and 20 mass%) composites at three different constant ramp rates of 20, 25, and 30°C/min in nitrogen gas atmosphere. The material was heated to a temperature of around 5–6 mg. High thermal stability is the capacity to sustain necessary mechanical qualities at a specific temperature,

such as strength, toughness, and elasticity. The non-isothermal crystallization technique was carried out at a variable temperature suitable for industrial processes and simple to carry out in research. The melting behavior owing to non-isothermal crystallization was studied using TGA.

2.4.4 | Water contact angle measurement

The water contact angle (WCA) was employed using static sessile drop method at room temperature using doubled distilled water (DMS-401, Kyowa Interface Science Co. Ltd.). The 2 μL liquid droplet was dropped onto the sample surface. After 3 s of liquid deposition, a charge coupled device (CCD) type camera was employed to record an image of the static WCA over the sample's surface after the syringe was removed. The software inbuilt with the CA measuring machine was used to calculate the WCA. The calculated WCA values were determined on the triplicate samples by aggregating four successive measurements, which were always taken on the samples' previously unsaturated areas.

2.4.5 | Hardness test

Hardness test was conducted at room temperature using a digital Rockwell hardness tester (RASNE-3, Fuel Instruments & Engineering Pvt. Ltd.). The hardness of the sample was measured using the “F” scale, with the tip diameter of 1/16th inch steel ball indenter setting the load adjusted at 60 kgf, and the dwell time for 10 s.

2.4.6 | Tensile study

In this study, a tensile test was conducted at normal atmosphere at a crosshead speed of 2 mm/min using a bench-type Tensometer (A370-E8, Kudale Instruments Pvt. Ltd.). From the stress versus strain plots, the Young's modulus, ultimate tensile strength (UTS), and strain were evaluated.

2.4.7 | Tribological test

The tribological test was performed with ASTM G99 method using a tribometer (TR-201, DUCO M tribometer, Docom Instrument, India) under the dry condition, where the pin was in static condition and disc was in rotational condition. The testing machine was placed on the table, which holds the rotational disc, which is

driven by servomotor. The upper pivoting arm was fixed to a frame of the machine through bearings to apply a normal load (N) to the rotating disc through the pin specimen using a deadweight loading system. Prior to test, the pin specimens were polished against 600, 1000, 1200, and 2000-grade SiC emery papers to guarantee a good contact between the sliding pin surface and the counterface of the disc during the dry sliding experiment. The wear testing condition and constrain of PA6 thermoplastic and PA6/GF (i.e., 5, 10, and 20 mass %) composites are shown in Table 2. It would clarify the tribological behavior of neat PA6 and GF-reinforced PA6 composite pins sliding against EN31 steel rotating disc under dry sliding condition, with different normal load (viz., 10, 20, and 30 N), sliding velocity (viz., 200, 400, and 800 rpm), and run timing (viz., 30, 45, and 60 min) in order to provide some useful road maps for employing such composites in dry sliding condition.

During the wear test, the COF for each sample was measured and consequently, the specific (WR, mm³/Nm) was determined following Equation (1).^{31,32}

$$WR = \frac{\Delta m \text{ mm}^3}{\rho F d \text{ Nm}}, \quad (1)$$

where Δm = mass loss (g) of the sample measured using a weighing balance of 5 decimal digits, ρ = density of the samples (g/mm³), F = normal load (N), and d = sliding distance (m). In order to calculate an exact quantity of mass loss, the specimens were dried out using normal atmosphere for a period of 24 h before the mass loss was measured.

2.4.8 | Optical microscopic study

The Olympus BX41M-LED digital microscope was used to evaluate the worn surface region, subsurface, worn debris, and morphology of wear and tensile test specimens at room temperature. The morphology of pure PA6 and composites with GF (5, 10, and 20 mass%) reinforcement was investigated using optical microscopy.

2.4.9 | Scanning electron microscopy and elemental study

After the wear test, the morphology of the specimen's worn surfaces was examined by scanning electron microscopy using a SNE3200M-SEC scanning electron microscope (SEM) and an energy dispersive spectrometer (EDS) to analyze the surface morphology and chemical composition of PA6, respectively, of the GF reinforced PA6 composites.

2.4.10 | Design of experiment

In this study, MINITAB 16 was used to analyze the tribological results. The Taguchi approach employed the design of experiments (DOE) to obtain the most significant data with the least amount of experiments.

TABLE 2 Constrain of PA6 polymer and PA6/GF(5, 10, and 20 mass %) composites for wear test.

Constrain	Testing condition
Load	10, 20, and 30 N
Speed	200, 400, and 800 rpm
Time	30, 45, and 60 min
Pin materials	PA6 without GF and PA6 with GF (i.e., 5, 10, and 20 mass%) composites
Surface condition	Dry friction
Track diameter	80 mm
Disc material	EN31 steel
Temperature	25°C
Relative humidity	40%–45%

Abbreviation: GF, glass fiber.

TABLE 3 Process parameters and their levels for the composite systems PA6/GF(0, 5, and 10) and PA6/GF(0, 5, and 20).

Factors Process parameters	Levels for composites PA6/GF(0, 5, and 10)			Levels for composites PA6/GF(0, 5, and 20)		
	1	2	3	1	2	3
GF mass%	0	5	10	0	5	20
Time (S)	30	45	60	30	45	60
Load (N)	10	20	30	10	20	30
Speed (rpm)	200	400	800	200	400	800

Abbreviation: GF, glass fiber.

According to the Taguchi method's L27 orthogonal array (OA), the experimental design was set up. The control factor process parameters and the levels are shown in Table 3 for the PA6/GF composites (0, 5, and 10 mass% GF) and the PA6/GF composites (0, 5, and 20 mass% GF). In the present Taguchi optimization technique, to minimize the quality loss utilizing, the objective function was considered as “the-smaller-the-better” for both COF and WR output responses. It is to be informed that the according to the analysis in MINITAB, the highest “signal-to-noise” (S/N) ratio often yields the optimum quality with minimum variance irrespective to these objective functions. Each experiment was conducted twice to minimize the impact of independent variables (noise factors) on the results. The S/N ratio (η) was computed from the quality ratings using Equation (2).³³

$$\eta = -10 \log \frac{1}{n} \sum_{i=1}^n y^2, \quad (2)$$

where “ n ” = number experiment, “ y ” = number of response value, and “ η ” = output response.

Additionally, a statistical analysis of variance (ANOVA) was performed in order to find the process factors, which were statistically significant. The ideal mixture of the process parameters can be predicted with a respectable accuracy level using the S/N and ANOVA investigations. To evaluate the primary interaction factors statistically significant at the 95% confidence level, an ANOVA was conducted. The sum of squares (SS), F -ratio, and p -value columns constitute the ANOVA table. The first and second order terms, and the lack of fit and experimental error were used to compute the SS. The average squared error was divided by the sum of the squares of the factors after correction to get the F -ratio. If the p -value is less than or equal to 0.05, the factors are considered statistically significant according to the ANOVA. The p -value shows the likelihood that each variable will have an impact on the thrust force (N), diameter (mm), and circularity error (μm).

3 | RESULTS AND DISCUSSION

3.1 | FTIR analysis

The FTIR spectra of raw GF, raw PA6 powder, PA6 plastic, and its composites with varying GF of 5, 10, and 20 mass% are shown in Figure 4a. Identification of polymers and its chemical structure have been analyzed with the help of FTIR spectroscopy.^{34,35} The complete band assignment and the molecular structure of the PA6 polymer were confirmed by FTIR data.³⁶ The residue spectra

of PA6 raw powder, PA6, PA6/5GF, PA6/10GF, and PA6/20GF were quite similar and there was no significant difference in the spectra of PA6 and GF-reinforced PA6 composites.^{37,38} More precise variations of FTIR peaks of those composites at specific wavenumber ranges are depicted in Figures 2b and 4g. The peaks of the amide groups of 1540 cm^{-1} (amide I band) as well as 1635 cm^{-1} (amide II/CH₂ asymmetries) were identified. In addition, bands at 3296 cm^{-1} for N—H stretching, 2930 cm^{-1} for CH₂ asymmetric stretching, and 2860 cm^{-1} CH₂ symmetric stretching, 689 cm^{-1} for C—C deformation bend, and 575 cm^{-1} for O=C—N bending assigned to PA6 molecules were noticed. Besides all the pure PA6's functional groups, some important peaks of GF also noticed nearly at 970 , 1174 , and 1260 cm^{-1} assigned to Si—OH, Si—O—Si, and ester carbonyl C=O of silanized GF. It has been shown that the bands at 1170 cm^{-1} , which is functional group of CO—NH and skeletal motion, represent the amorphous phase of PA6.³⁹ Table 4 illustrates the FTIR peaks of the specific functional groups for the GF, PA6, and PA6/GF composites.

3.2 | XRD analysis

The XRD patterns of PA6 raw powder, pure PA6, PA6/5GF, PA6/10GF, and PA6/20GF composites, as well as GF powder, are illustrated in Figure 3. There are two types of crystalline structures in PA6: pseudo-hexagonal crystallites (γ phase) corresponding to the peaks $\gamma(001)$ at $2\theta = 20.1^\circ$ and $\gamma(002)$ at $2\theta = 21.8^\circ$ and monoclinic structure (α phase) attributed to the peak $\alpha(002)$ at $2\theta = 23.8^\circ$.^{40–42} It can be seen that for raw PA6 powder and pure PA6 plastic pin, these three main peaks did not change much. However, after addition of GFs, which was mainly identified by silica peak SiO₂(210) at $2\theta = 28.3^\circ$,⁴³ to the PA6 system, the $\alpha(002)$ had grown significantly compared to the $\gamma(001)$ phase. This result indicates that the melting point of PA6 matrix would increase for the composites due to the formation of predominant monoclinic α phase in composite and it will be determined by further DTA. A similar result also reported by Majka et al., 2018 where the pseudo-hexagonal crystallites (γ phase) melted at 215°C and the monoclinic structure (α phase) melted at 224°C .⁴² In addition, there was another small peak found in GF reinforced composites corresponding to the γ phase Si (111) at $2\theta = 27.64^\circ$,^{44,45} which might be the cause of thermal history of material.

3.3 | TGA and activation energy (E_a)

Figure 4 depicts the results of TGA used to investigate the mass loss and thermal stability of pure PA6 and

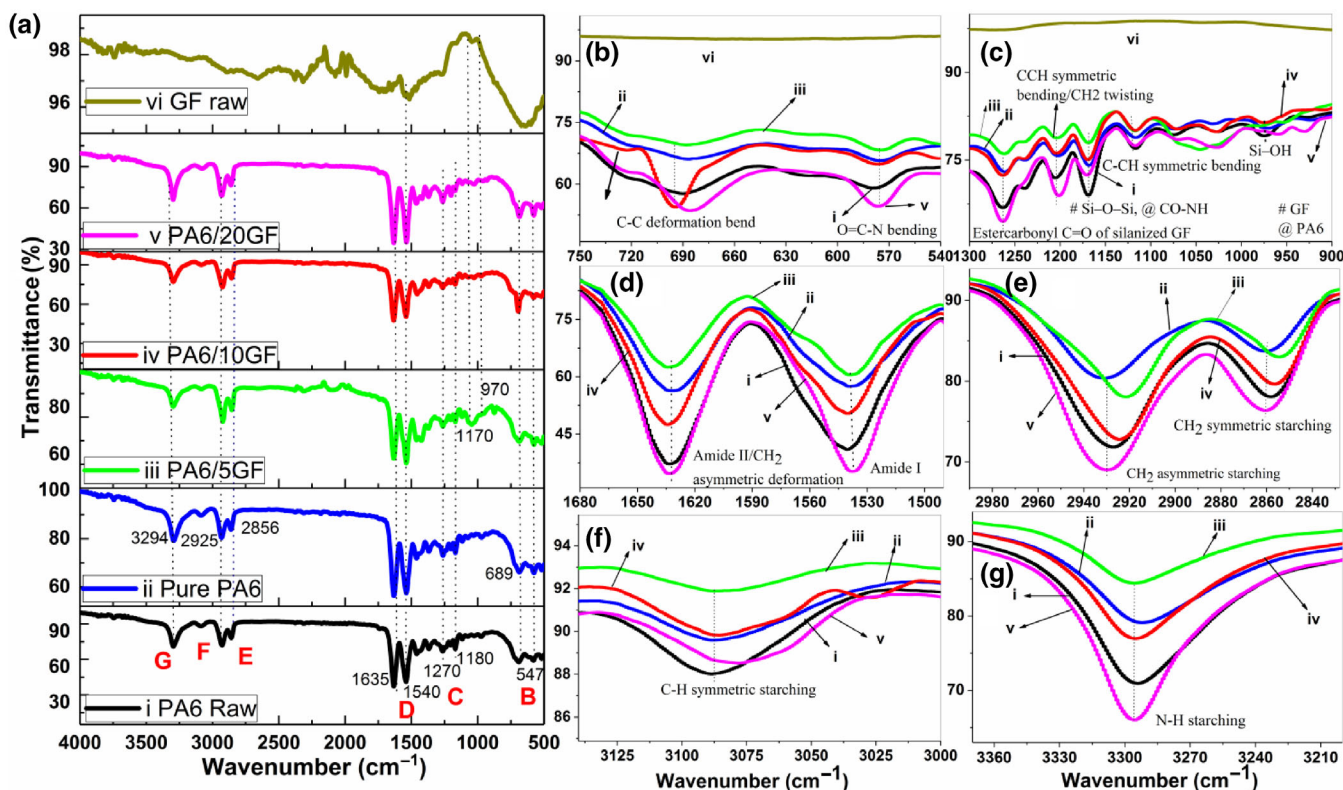


FIGURE 2 (a) Full range FTIR spectra for PA6 powder, GF raw, and PA6, PA6/5GF, PA6/10GF, and PA6/5GF/20GF composites, and the precise variations of FTIR peaks of those composites at specific wavenumber ranges of (b) 815–490 cm^{-1} , (c) 1400–1000 cm^{-1} , (d) 1830–1310 cm^{-1} , (e) 3000–2700 cm^{-1} , (f) 3230–2975 cm^{-1} , and (g) 3400–3150 cm^{-1} . FTIR, Fourier transformation infrared; GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 4 FTIR spectra of PA6 powder, and PA6, PA6/5GF, PA6/10GF, and PA6/20GF composites.

Functional group	PA6 cm^{-1}	PA6/5GF cm^{-1}	PA6/10GF cm^{-1}	PA6/20GF cm^{-1}
N–H stretching	3296	3295	3294	3294
C–H symmetric stretching	3020	3020	3020	3020
CH ₂ asymmetric stretching	2930	2920	2923	2929
CH ₂ symmetric stretching	2860	2854	2854	2861
Amide II/CH ₂ asymmetric deformation	1635	1632	1635	1635
Amide I	1540	1568	1540	1538
Ester carbonyl C=O of silanized GF	1260	1260	1260	1260
CCH symmetric bending/CH ₂ twisting	1205	1207	1202	1202
Si–O–Si	–	1174	1174	1174
CO–NH, skeletal motion	1180	1180	1180	1180
CCH symmetric bending	1128	1128	1128	1128
Si–OH	–	970	970	970
C–C deformation bend	689	689	689	689
O=C–N bending	575	575	573	573

Abbreviations: FTIR, Fourier transformation infrared; GF, glass fiber.

PA6/GF(5, 10, and 20 mass%) composite samples. The weight loss curves for thermal degradation of pure PA6 and GF reinforced PA6 composites in a continuous mode

throughout a range of temperatures were at three different heating rates of 20, 25, and 30°C/min. The TGA results show that the GF is thermally stable up to the

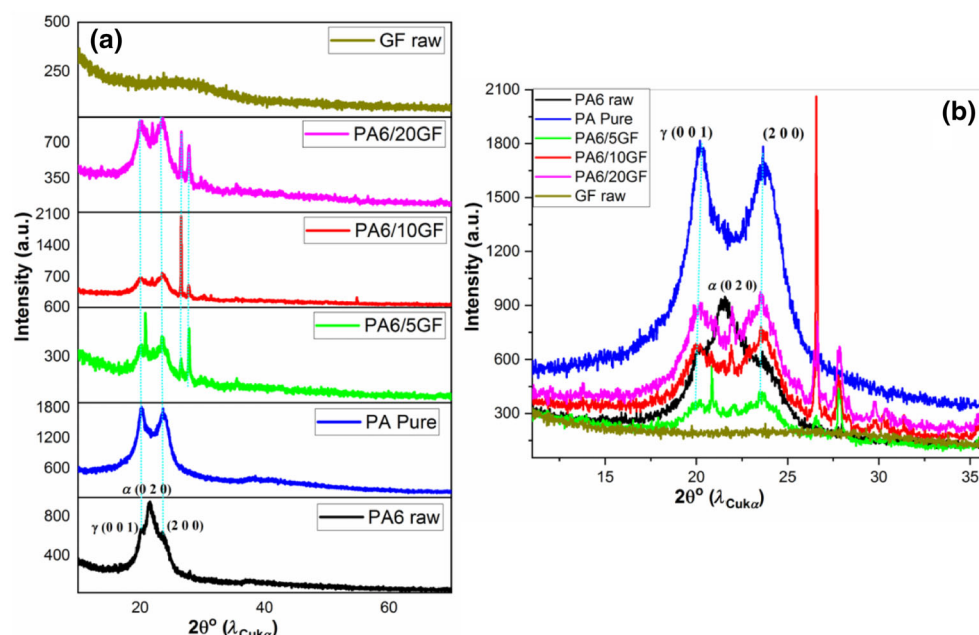


FIGURE 3 (a) Full range XRD spectra and (b) peak for PA6 raw powder, chopped GF, and pure PA6, PA6/5GF, PA6/10GF and PA6/5GF/5Grt composites. GF, glass fiber, XRD, x-ray diffractometer. [Color figure can be viewed at wileyonlinelibrary.com]

maximum tested temperature 800°C. Moreover, the maximum degradation onset temperature was found 404°C for PA6/5GF. Further reductions were obtained at 399 and 324°C for PA6/10GF and PA6/20GF, respectively. The pure PA6 degrades at a lower onset temperature than that of GF reinforced PA6/5GF composite, which was caused by the influence of high-temperature GF reinforced in PA6 composites up to a certain limit.⁴⁶ It also caused by the dominant monoclinic α phase formation in the composites, as determined by the XRD analysis. One of the most essential parameters in the process of phase transition is the activation energy, which is connected to the energy and potential barrier. It is a significant indication in determining the ability of a polymer since it may be used to reflect the amount of energy that is necessary to overcome the potential barrier in polymer.⁴⁷ The activation energy (E_a) for the conversion between 300 and 500°C temperatures calculated using Coats and Redfern Equation (3) is illustrated in Table 5.^{48,49}

$$\log \left[\frac{-\log(1-\alpha)}{T^2} \right] = \log \frac{AR}{\beta E_a} \left[1 - \frac{2RT}{E_a} \right] - \frac{E_a}{2.303RT}, \quad (3)$$

where α is the fraction of sample decomposed at time (t) can be calculated using Equation (4):

$$\alpha = \frac{W_0 - W_t}{W_0 - W_f}, \quad (4)$$

where W_0 represents the sample initial weight (previous to the decomposition reaction occurring), W_t stands for the sample weight at a temperature of 300–500°C, and W_f represents the sample final weight (after the reaction has been completed). Additionally, T represents the sample

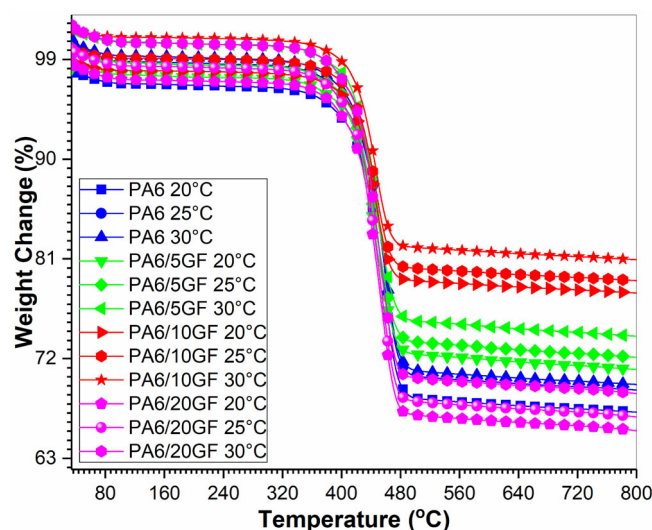


FIGURE 4 TGA for Pure PA6, PA6/5GF, PA6/10GF, PA6/20GF composites material at 20, 25, and 30°C/min heating rates. TGA, thermogravimetric analysis. [Color figure can be viewed at wileyonlinelibrary.com]

absolute temperature, A represents the frequency factor, and R represents the gas constant (8.314×10^3 J/K/mol).

The degradation behavior of the samples and the related activation energies have a relationship with the E_a , which is required to start the thermal degradation process, which increases with the increase of GF content in the composites. The plots between $\log \left[\frac{-\log(1-\alpha)}{T^2} \right]$ versus $\frac{1000}{T}$ at the heating rates of 20, 25, and 30°C/min are illustrated in Figure 5. From the Table 5, it has also been found that the activation energy (E_a) of the thermal decomposition of the samples increases with addition of

TABLE 5 Thermal decomposition at different mass fraction, slope and activation energy (E_a) obtained from TGA thermograms of pure PA6 and PA6/5GF, PA6/10GF, PA6/20GF composites material with 20, 25, and 30°C/min heating rates.

Samples with different heating rates	Temperature at different mass losses, °C					Activation energy	
	T_0	T_5	T_{10}	T_{20}	$T_{\max}$	Slope	E_a
	At 0 mass%	At 5 mass%	At 10 mass%	At 20 mass%	At maximum mass%		(300–500°C) kJ/mol
PA6 at 20°C/min	336.98	418.54	438.53	460.56	484.12	−8.710	166.771
PA6 at 25°C/min	353.02	425.24	442.90	462.84	486.10	−9.676	185.267
PA6 at 30°C/min	362.48	416.15	444.35	460.65	485.05	−9.925	190.003
PA6/5GF 20°C/min	340.95	415.10	433.98	458.46	479.27	−9.843	188.465
PA6/5GF 25°C/min	349.69	420.69	437.94	462.93	488.41	−10.507	201.179
PA6/5GF 30°C/min	355.29	421.92	438.52	464.06	489.59	−11.700	224.021
PA6/10GF 20°C/min	350.98	415.27	429.61	451.82	482.77	−12.730	243.743
PA6/10GF 25°C/min	356.29	416.44	433.98	456.19	480.67	−12.910	247.189
PA6/10GF 30°C/min	357.56	417.37	430.66	449.55	481.72	−13.395	256.475
PA6/20GF 20°C/min	360.88	412.82	431.71	453.92	483.82	−12.435	238.094
PA6/20GF 25°C/min	371.90	411.95	428.39	449.37	484.34	−12.668	242.555
PA6/20GF 30°C/min	375.22	415.10	432.93	448.50	489.59	−12.515	239.626

Abbreviations: GF, glass fiber; TGA, thermogravimetric analysis.

GF content in PA6. It also observed that the E_a increases with the addition of the GF reinforcement compare to pure PA6 material and maximum E_a are found to be 10 mass % GF into PA6 matrix this may be the good interfacial bonding between the matrix and reinforcement. Furthermore, it is notable that the E_a decreases when a 20 mass% GF reinforcement is included into the PA6 matrix. This may be the weak connection between the GF reinforcement and the PA6 matrix.

The Kissinger technique, which was created by Homer E. Kissinger in 1957, is an additional way that is used to determine the activation energy (E_a). This method is very quick and simple to use, it has emerged as one of the most often used method for finding kinetic parameters via the utilization of TGA data. The Kissinger is used to the E_a without the need for prior knowledge of the reaction process with the help of slope of resulting straight line.⁵⁰ Kissinger method was approached through the Equation (5).⁵¹

$$\ln \left[\frac{q}{T_p^2} \right] = \ln \left[\frac{AR}{E_a} \right] - \frac{E_a}{RT_p}, \quad (5)$$

where q represents the rate of heating, T_p represents the temperature at the point where the TGA curve is influenced, and the plot of the $\ln \left[\frac{q}{T_p^2} \right]$ versus $-\left[\frac{1}{T_p} \right]$ provides a straight line. The E_a may be estimated by calculating the slope of the line, which is denoted as E_a/R .

The E_a was calculated using Equation (5), which is used to plot the graph of $\ln \left[\frac{q}{T_p^2} \right]$ versus $-\left[\frac{1}{T_p} \right]$ in Figure 6. The slope of the line for PA6 and GF filled with PA6 composites is also used to determine the E_a , which is shown in Table 6. The findings reveal that there is a direct relationship between the E_a and increasing GF content with the consistent TGA profile. The E_a of the composites shows a positive correlation with the increasing GF content. This finding aligns with the findings obtained using the Kissinger approach and suggests that reinforcement functions as a thermal barrier, effectively retarding the process of deterioration. Additionally, it is worth mentioning that the E_a is decreased when 20 mass % GF reinforcement is added to the PA6 matrix. This might possibly clarify the weak bonding between the GF reinforcement and the PA6 matrix. The E_a of the PA6 materials with different GF content (5, 10, and 20 GF mass%) increases according to the following order: pure PA6 < PA6/20GF < PA6/5GF < PA6/10GF. From the Tables 6 and 7, it can be concluded that the PA6 with GF filled has high decomposition E_a compared to unfilled GF PA6 material.

3.4 | WCA measurement

PA6 is well known to be hygroscopic material and absorbs water easily even at room temperature.⁵² Figure 7 illustrates the WCA for PA6 and PA6/GF(5, 10, and 20 mass%) composites. The results indicate that the

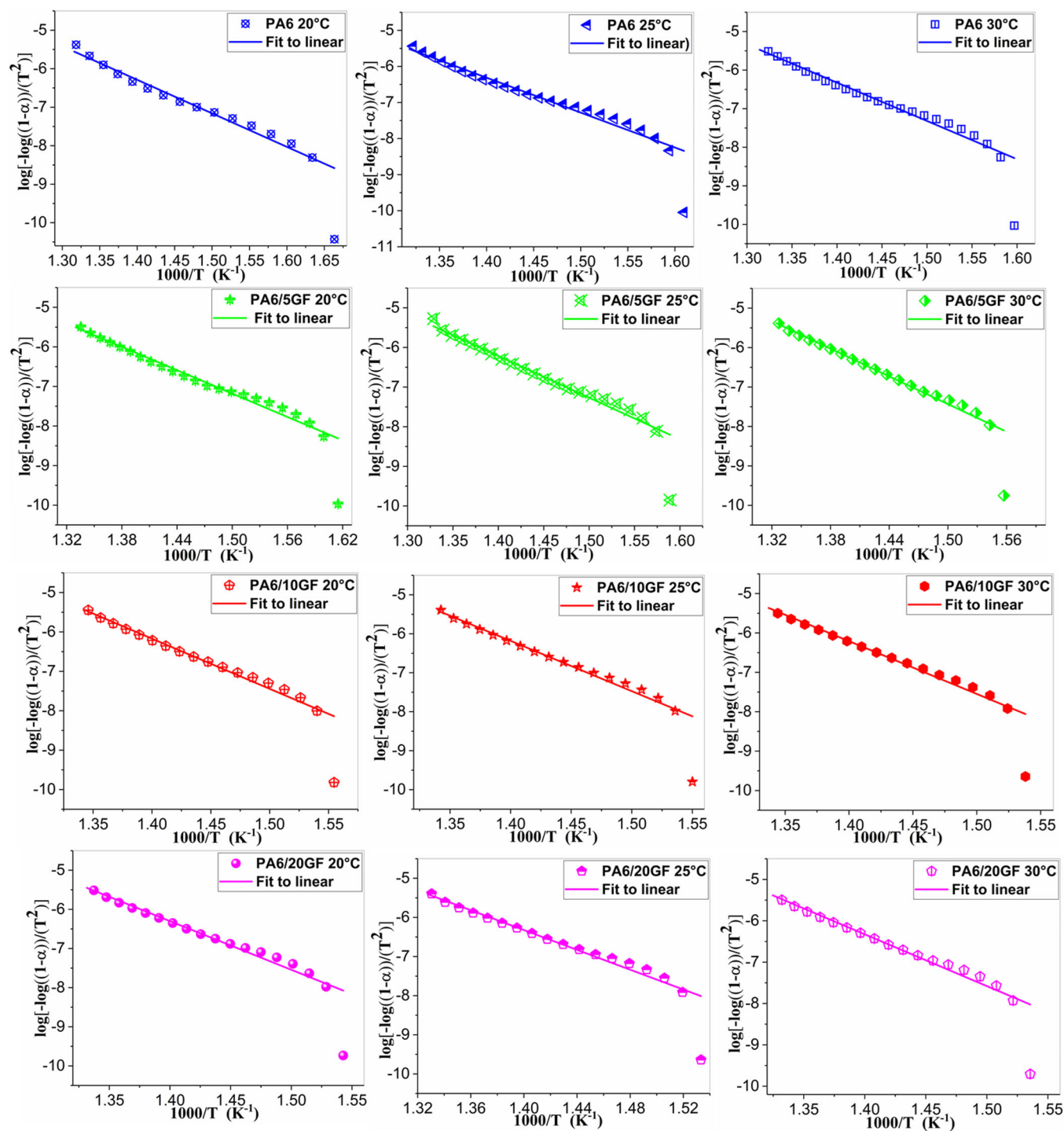


FIGURE 5 Plots of $\log\left[\frac{-\log(1-\alpha)}{T^2}\right]$ versus $\frac{1000}{T}$ for Pure PA6, PA6/5GF, PA6/10GF, PA6/20GF composites material at 20, 25, and 30°C heating rates. GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

average water wetting angle for PA6, PA6/5GF, PA6/10GF, and PA6/20GF composites is 87°, 111°, 103°, and 102°, respectively. The observed outcomes showed that PA6 is hydrophilic in nature and after adding GF, it tends to change into hydrophobic. The most hydrophobic property was found at 5 mass% GF for the PA6/GF composite but after adding 10 and 20 mass% GF, the hydrophobicity little reduced.⁵³

3.5 | Rockwell hardness test

The effect of GF on surface-mechanical properties of the PA6 and PA6/GF(5, 10, and 20 mass%) composites was studied through Rockwell hardness test in “F” scale and the results are shown in Figure 8a. The mean hardness of the PA6, PA6/5GF, PA6/10GF, and PA6/20GF composites found to be 40.57, 45.20, 40.52, and 38.68 HRF,

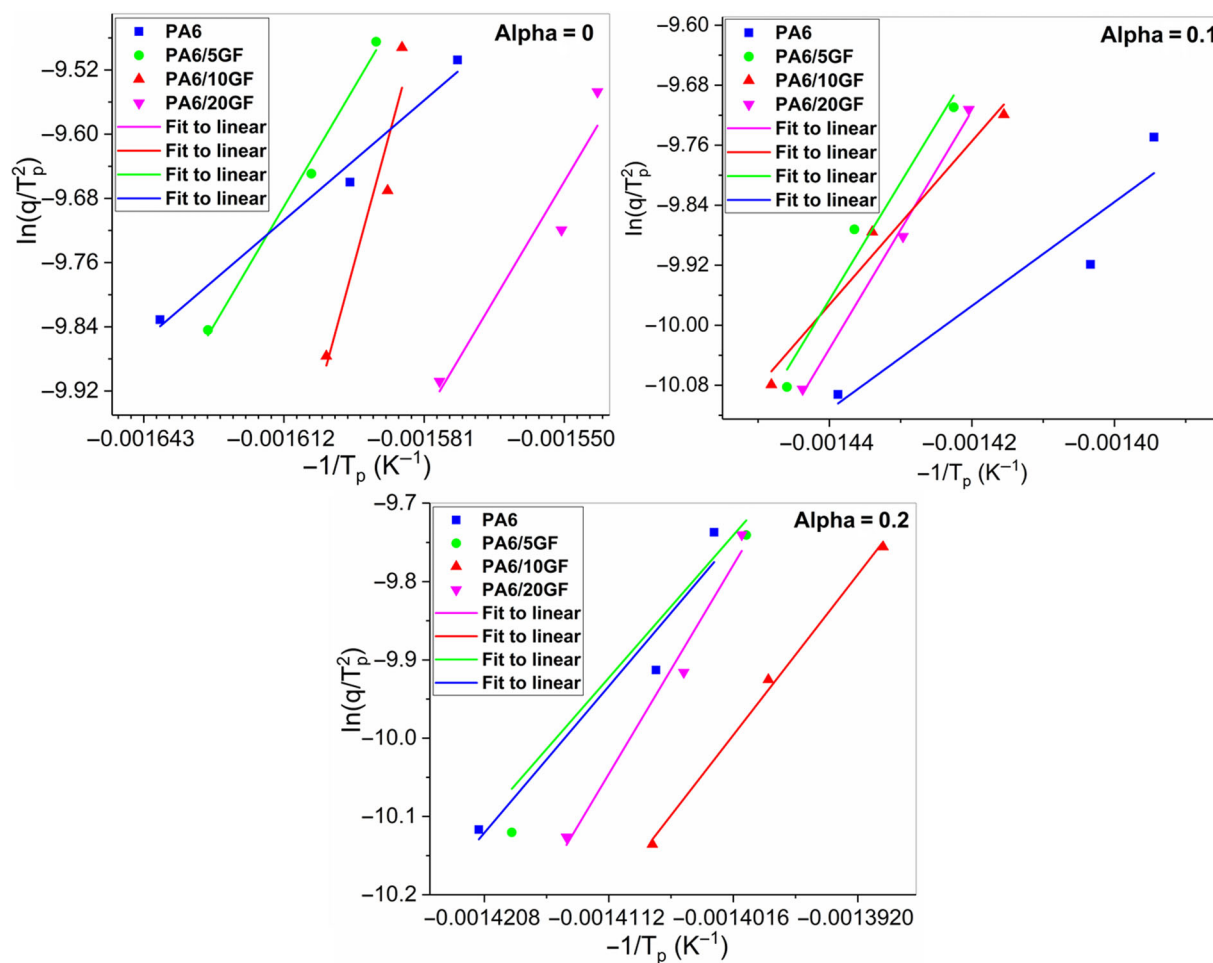


FIGURE 6 Plot of $\ln \left[\frac{q}{T_p^2} \right]$ versus $-\left[\frac{1}{T_p} \right]$ for Pure PA6, PA6/5GF, PA6/10GF, PA6/20GF composites material at 20, 25, and 30°C/min heating rates with different conversion degree by Kissinger method. GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 6 Slope details and activation energy (E_a) for Pure PA6 and PA6/5GF, PA6/10GF, PA6/20GF composites material with different conversion degree by Kissinger method.

Sample	Slope ($\alpha = 0$)	E_a kJ/mol	Slope ($\alpha = 0.1$)	E_a kJ/mol	Slope ($\alpha = 0.2$)	E_a kJ/mol
Pure PA6	4821.61	40.86	6922.91	57.56	19521.82	162.30
PA6/5GF	9559.01	79.47	15621.82	129.88	18973.53	157.7
PA6/10GF	20704.57	172.14	10900.98	90.63	21367.48	177.65
PA6/20GF	9484.01	78.85	15866.40	131.92	27805.92	231.18

Abbreviations: GF, glass fiber.

TABLE 7 Mechanical properties of PA6 and PA6/5GF(5, 10 and 20 mass%) composites.

Samples	Rockwell hardness (HR _F)	Elastic modulus (GPa)	Ultimate tensile strength, UTS (MPa)	Elongation at UTS (%)	Elongation at fracture (%)
PA6	40.57	0.21	32.14	20.6	70
PA6/5GF	45.20	0.39	44.46	16.4	25
PA6/10GF	40.52	0.37	43.70	30	18
PA6/20GF	38.68	0.32	43.50	23	28

Abbreviation: GF, glass fiber.

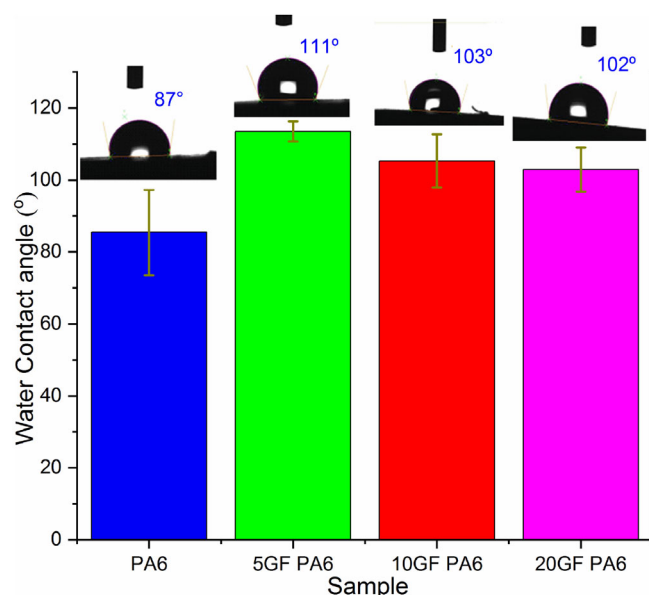


FIGURE 7 Variation of WCA for PA6, PA6/5GF, PA6/10GF, and PA6/20GF composites samples. GF, glass fiber; WCA, water contact angle. [Color figure can be viewed at wileyonlinelibrary.com]

respectively. It has been seen that the addition of GF into PA6 matrix enhanced surface hardness of the material. The addition of 5 mass% GF effectively enhanced the hardness at most and further addition of GF decreased the hardness drastically.

3.6 | Tensile test

The tensile test was done to determine the effect of GF on the bulk mechanical properties of PA6 composites. A design of tensile test specimens is shown in the inset of Figure 8a. The results indicate that PA6 has a low elastic modulus, whereas the addition of GF was able to enhance the elastic modulus of the composites. The comparison in tensile properties of PA6 with PA6/GF composites at 5, 10, and 20 mass% GF is illustrated in Figure 8b. The inset of Figure 8b shows the tensile test samples both before and after the test. According to the outcomes of the tensile test, the addition of 5 mass% GF to the PA6 matrix had attributed to the greatest improvement in its tensile properties including UTS of 44.46 MPa and elongation up to 25%. The present UTS and elongation are significantly higher than the other composites made of like, GF reinforced PA6/acrylonitrile-butadiene-styrene with 39 MPa and 18% elongation⁵⁴ or GF reinforced PA6/ polypropylene with 35.23 MPa and nearly 5% elongation.⁵⁵ On the other hand, adding

10 and 20 mass% GF caused a reduction in the material's tensile strengths. It might be the cause of agglomeration of GFs during mixing with PA6 matrix. The elastic modulus, UTS, elongation at UTS, and elongation at fracture are summarized in Table 7.

Tensile fracture surfaces of PA6 and GF filled PA6 composites examined the optical microscopic are depicted in Figure 9. The most suitable fracture mechanisms have been identified by the fractographics study. The identified fracture mechanisms can be classified as classical mechanism presented by crack propagation, layer de-bonding, fiber breakage, fiber pull out, deflection, and ply breakage.⁵⁶ In the Figure 9a, it clearly reveals that a high plastic deformation and some broken polymer bundles appeared on the surface of the pure PA6 due to the absence of GF reinforcement. In contrast, Figure 9b reveals the good interfacial adhere bonding between GF and PA6 matrix, which resulted to the best tensile strength. Hence, less amount of plastic deformation was found on the 5 mass% GF filled PA6 composites' surface. Further increasing of GF content as 10 and 20 mass% in the composites (see Figure 9c,d), fiber pull out occurred due to inadequate fiber-matrix bonding observed might be the effect of agglomeration of tiny chopped fibers. Especially, the PA6/10GF sample showed smooth clean surface indicating the fiber and matrix would have very brittle fracture surface. Thus, mechanical properties again deteriorated and the values are mentioned in Table 7.

3.7 | Friction properties for PA6 and PA6/GF composites

To determine the effect of GF on wear behaviors of PA6 and PA6/GF(5, 10, and 20 mass%) based composites, the dry wear test was conducted by using rotating disc with different run times, normal loads, and sliding velocities. It can be noticed from Figures 10–13 that the COF depends on the parameters of load (in N), sliding velocity (in rpm), run time (in min), and content of GF.⁵⁷ However, it was noted that the composite material exhibits an emergence in coefficient of friction with variations in load and velocity during the experiment. In addition, COF was found to be increased rapidly in initial and then, became stable at end of the test. The reason for the rapid increase in COF might be due to the differences in the surface roughness.

From the Figures 10 and 11, it has been observed that the COF decreased with raising of normal load, run time, and sliding speed of pure PA6 and 5 mass% GF reinforced PA6 composite.¹⁵ It is widely known that the COF of

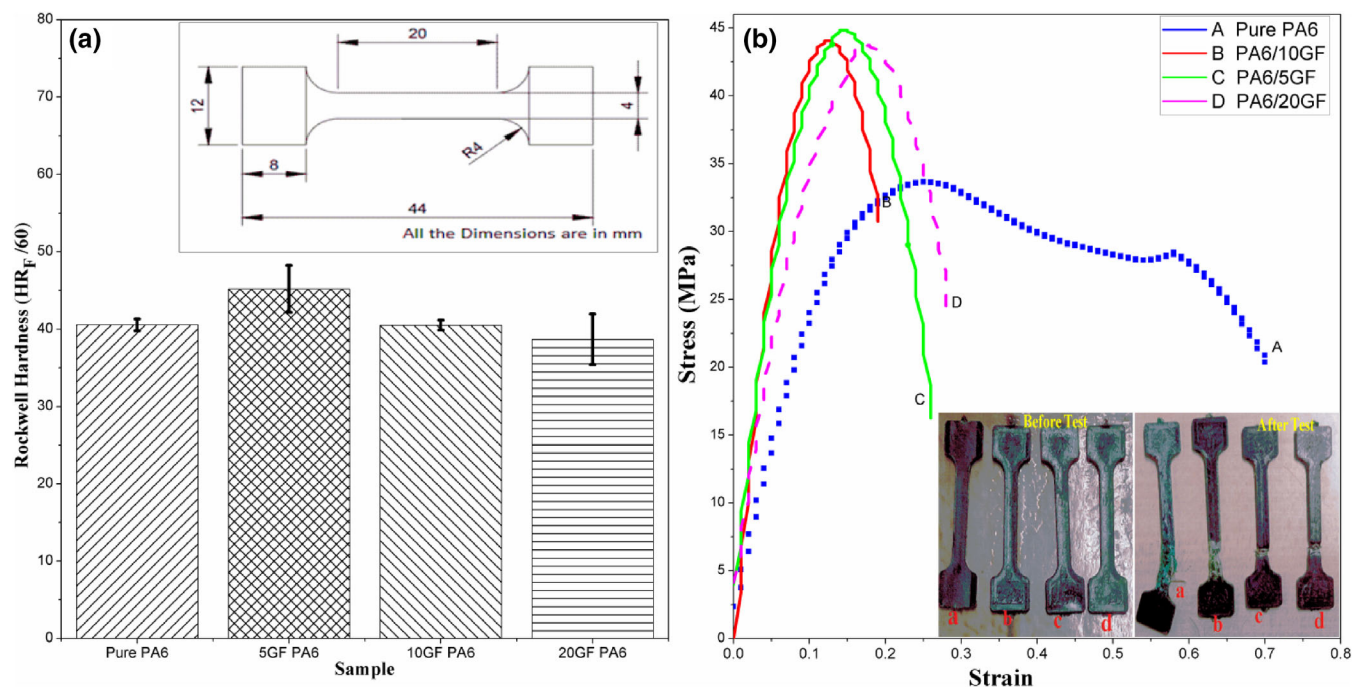
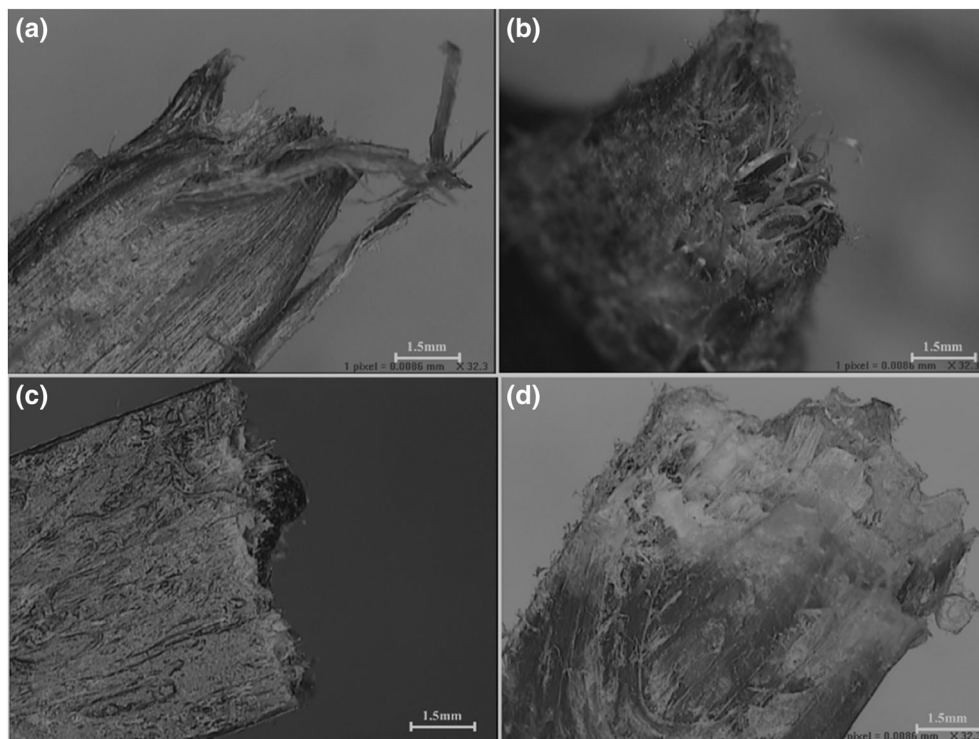


FIGURE 8 (a) Rockwell hardness and (b) stress–strain curve for PA6, and PA6/5GF, PA6/10GF, PA6/20GF composite samples. GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

FIGURE 9 Fracture region of tensile test specimens (a) PA6, and (b) PA6/5GF, (c) PA6/10GF, and (d) PA6/20GF composites. GF, glass fiber.



polymers being a function of adhesion can make same product of the real contact area and the polymer's shear strength.⁵⁸ Due to the heat generated by friction, the contact interfacial heat between the polymeric composite pin and the counter-face generally rises as the load increases.

This causes two effects that compete with each other to have an effect on the COF. On the one hand, the COF reduced as a consequence of the matrix material's decreased shear resistance and as the matrix softens at higher temperatures, the real contact area increased,

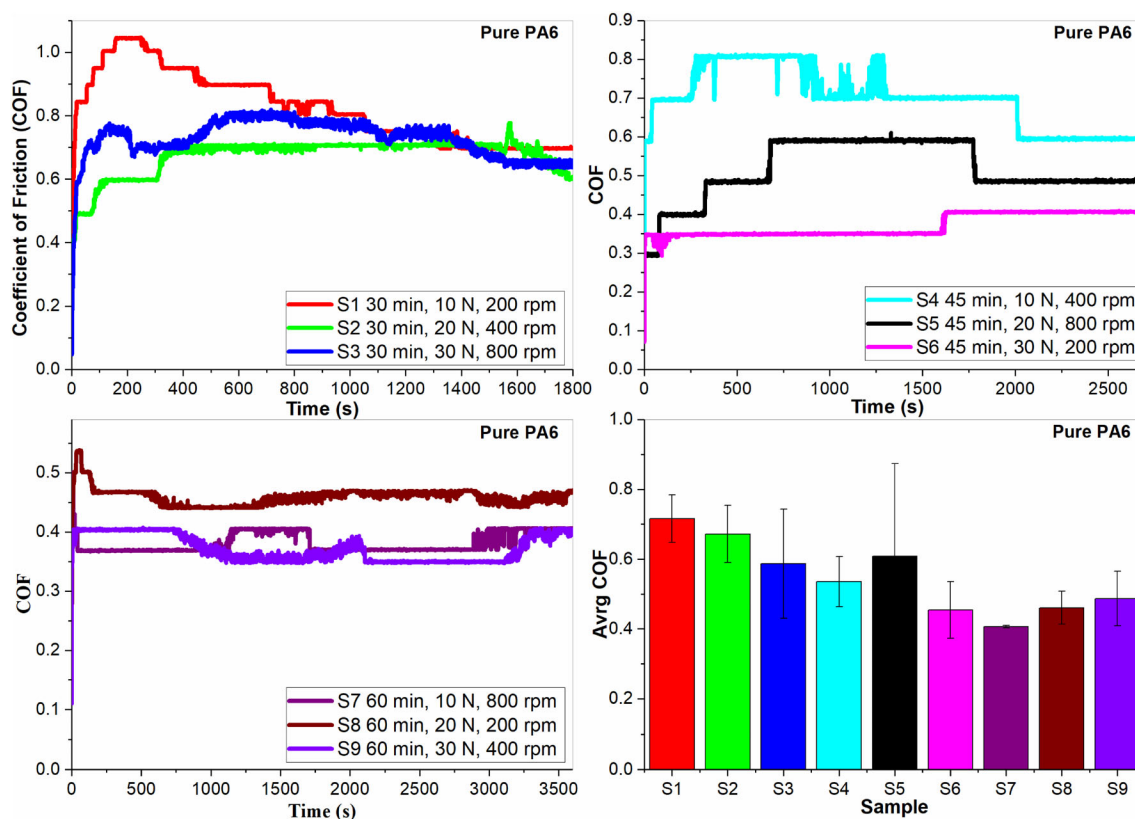


FIGURE 10 COF for PA6 pins at different run times, normal loads, and sliding velocities. COF, coefficient of friction. [Color figure can be viewed at wileyonlinelibrary.com]

which resulting to the increase in COF. Therefore, these two competing factors would influence the final COF of the polymer. From the Figures 12 and 13, it has been noticed that the COF was grown up with the increased normal load, sliding velocity and run time. Taking all the results into an account, the 5 mass% GF reinforced PA6 composites exhibited lowest COF among all tested composites in this research. This outcome may be an effect of a good interfacial adhere bonding between the matrix and fibers. The fibers of this PA6/GF composite did not detach easily from the matrix. The peeling process functioned as fiber wearing, thinning, and cracking, successively and thus, resulted in a reduction in the COF.^{59,60}

The average COF values are in the range of 0.27–0.33 for PA6 with 5 mass% GF (PA6/5GF), 0.39–0.30 for PA6/10GF, and 0.28–0.35 for PA6/20GF composites. However, the average COF is in the range of 0.48–0.71 for PA6. There is a significant reduction in the COF at PA6 with 5 mass% of GF content. Including all these results into consideration, it can be inferred that the PA6 with a GF 5 mass% content of displayed smallest COF among all the tested composites in this research work. This PA6/5GF composite showed almost similar COF for other composite system PK-25/PA6-50/GF30 developed by other research group.²²

3.8 | Wear properties for PA6 and PA6/GF composites

Figure 14 shows that the wear characteristics of PA6 and PA6/GF (5, 10, and 20 mass%) composites under normal load of 10, 20 and 30 N, with the sliding velocity of 200, 400, and 800 rpm and different run time 30, 45, and 60 min. The normal load (N) being one of the key elements influences the particular WR of composites. The WR of the composites material did not change significantly in pure PA6 material at any load.¹⁴ In contrast, the specific WR of PA6 with 5, 10, and 20 mass% of GF composites became lesser compared to pure PA6. This is due to the fact that the randomly oriented chopped GFs, which are firmly linked to the matrix PA6 material, protected the contact surfaces from various destructive processes by the counter face. This might have occurred as an outcome of the strong interfacial bonding properties, which were essential in transmitting loads from the matrix to hard fibrous particles.^{60,61} Moreover, GFs substantially improved the wear resistance behavior of the composites. From Figure 14, it has also been observed that the WR for all the PA6/GF composites decreased with increasing the normal load and sliding velocity.⁹ Specially, the

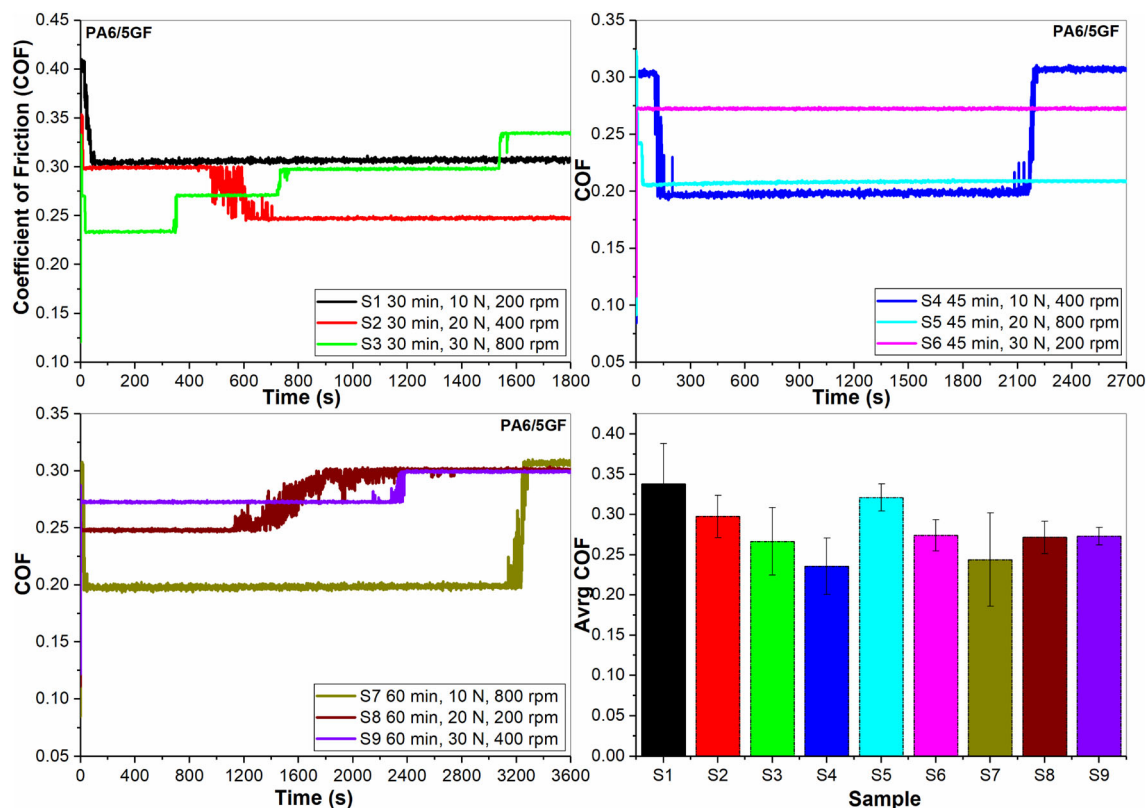


FIGURE 11 COF for PA6/5GF at different run times, normal loads, and sliding velocities. COF, coefficient of friction; GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

WR for PA6/5GF composites increased remarkably and even larger than that of pure PA6, but further addition of GF content as 10 and 20 mass% to PA6 did not affect in the improvement of the tribological properties due to the abrasive nature of broken GFs.⁶² This outcome was obtained owing to the weak interfacial adhesion between fiber and matrix. Furthermore, the micro-level scraping as well as cutting mechanisms that occur in matrix of the composite materials cannot be maintained and protected by the fibers. The obtained average WR was in the range of 5.7×10^{-6} – 1.2×10^{-6} mm³/Nm for PA6/5 mass% GF composite pins in comparisons with 7.4×10^{-6} – 1.4×10^{-6} mm³/Nm for pure PA6/10 mass% GF composites, and 9.1×10^{-6} – 1.9×10^{-6} mm³/Nm for PA6/20 mass% GF composite pins. However, the average WR was in the range of 1.8×10^{-5} – 1.5×10^{-5} mm³/Nm for pure PA6 plastic pins. The range of specific WRs of the present study is significantly lower and better than that of obtained for other composites for example, PA6/25GF of 8.484×10^{-3} mm³/Nm by other research group,²¹ but similar to the PK-25/PA6-50/GF30 composite system of 1.15×10^{-6} mm³/Nm reported by another research group.²²

3.9 | Optical microscopic study on wear surface

The microstructure of polymer matrix and the fiber/matrix interface play a vital role to find the mechanical characteristics of the thermoplastic composites. Figure 15 shows that the optical images for worn specimen for PA6, PA6/5GF, PA6/10GF, and PA6/20GF composite pin samples at load of 10, 20, and 30 N with sliding speed 200, 400, and 800 rpm, and run time 30, 45, and 60 min. Plenty of micro-cracks, plastic deformation and groove formation occurred in pure PA6 and are depicted in Figure 15a–c. Figures 15g–i show the 10 mass% of GF reinforced PA6 composites of wear worn surface. Figure 15g indicates that some broken particle, plastic deformation, and micro-crack, while with raising the load and sliding speed, the plastic deformation and micro-crack formation were not much, as seen in Figure 15e,i. Figures 15j–l show that the 20 mass% of GF PA6 reinforced composites of wear worn surface. Figure 15j indicates that the broken particles, plastic deformation, and micro-crack. With increasing the load and sliding velocity the plastic deformation and much groove formation occurred, as found in Figure 15. Further

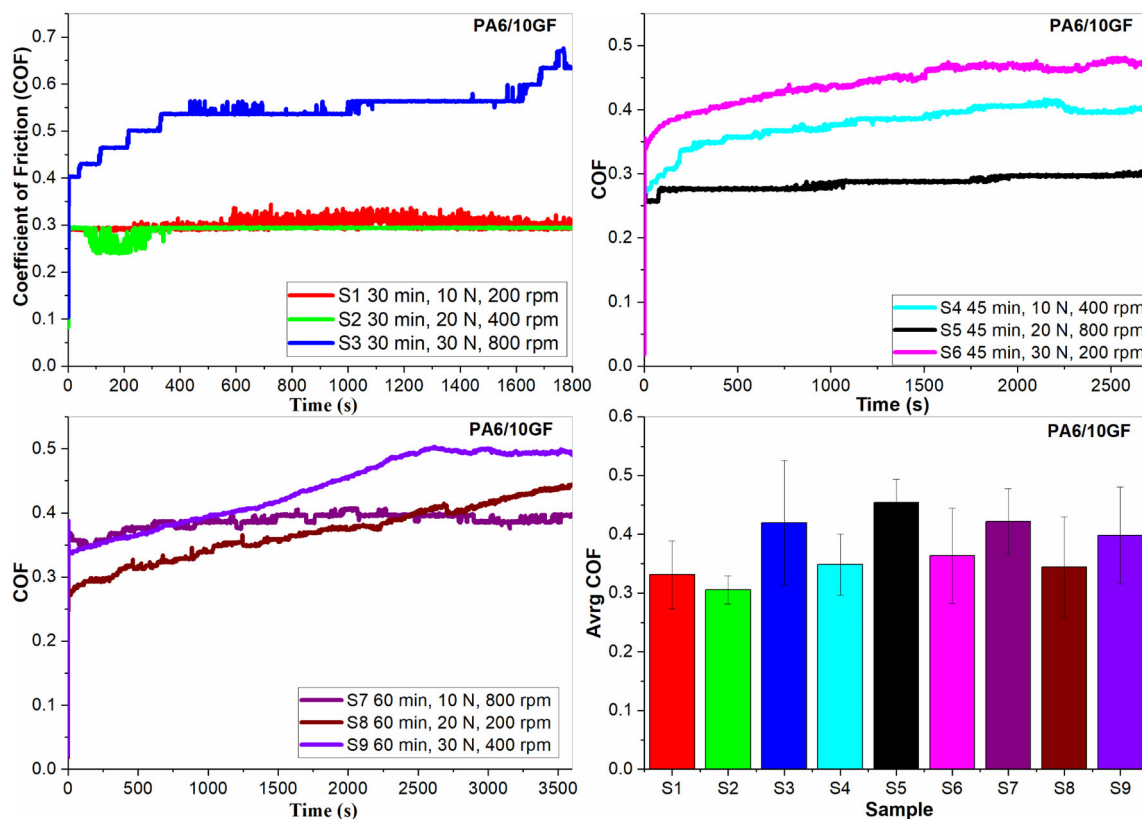


FIGURE 12 COF for PA6/10GF at different run times, normal loads, and sliding velocities. COF, coefficient of friction; GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

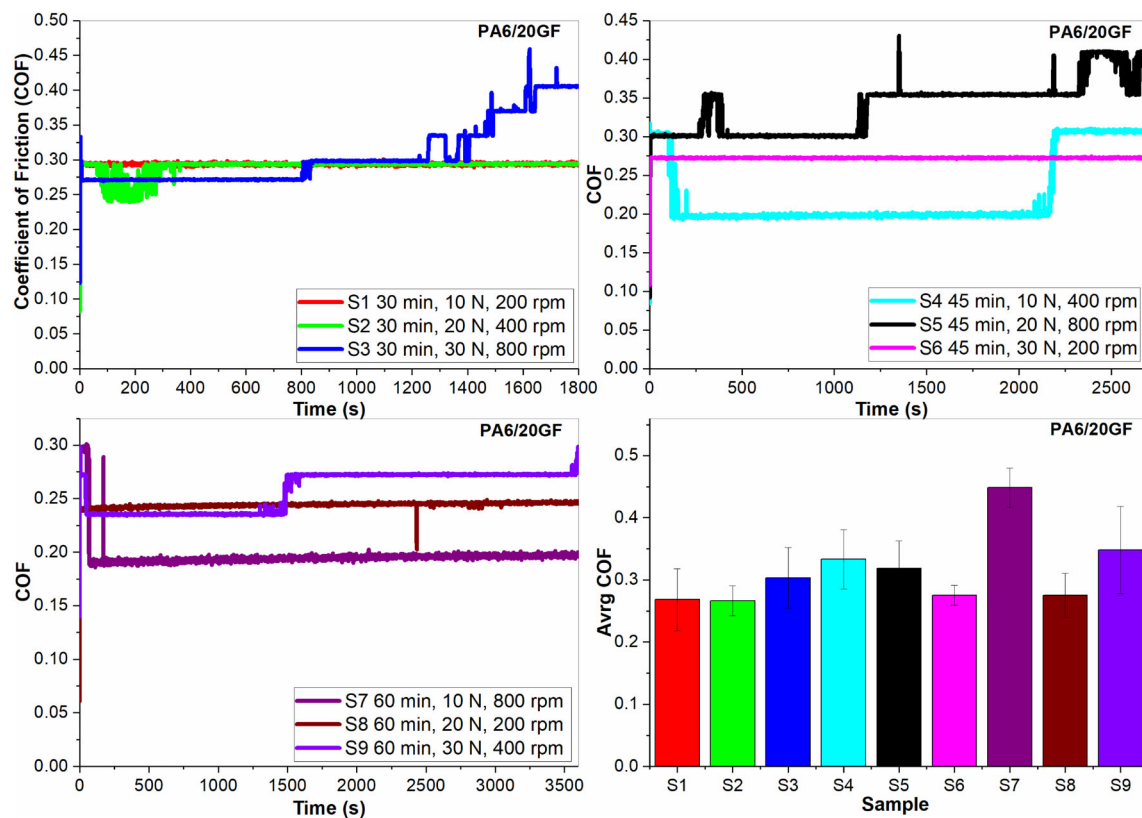


FIGURE 13 COF for PA6/20GF at different run times, normal loads, and sliding velocities. COF, coefficient of friction; GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

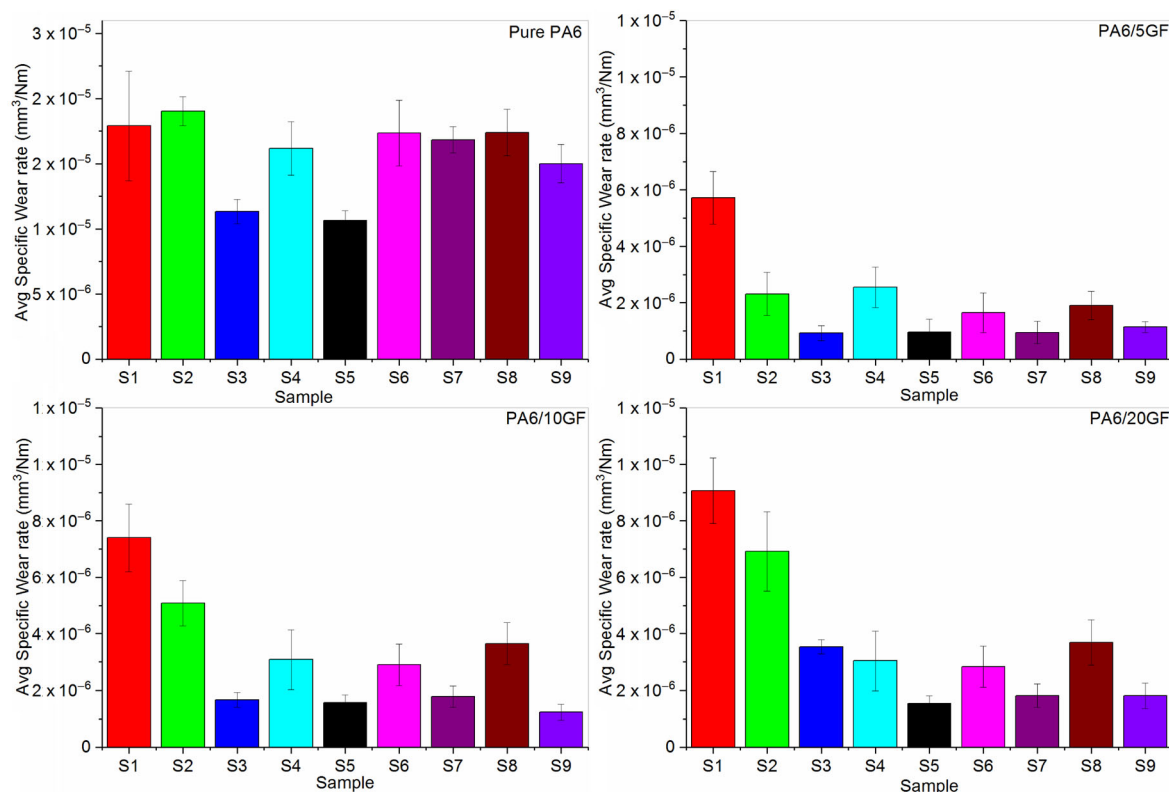


FIGURE 14 Average WR for PA6, PA6/5GF, PA6/10GF, and PA6/20GF at different run time, normal load, and sliding velocity. GF, glass fiber; WR, wear rate. [Color figure can be viewed at wileyonlinelibrary.com]

increasing the load and sliding velocity, the micro-crack formation was not generated significantly, as shown in Figure 15i. Furthermore, the matrix and particles in the PA6/5GF composite have been found to have strong interfacial adhesion in Figures 15d–f.

As a consequence, the debris holes, broken particles, and wedge pattern could not observe, as they are often found in the PA6/5GF composite specimens. These optical microstructures indicate that an increase in the concentration of GF in PA6 results to a significant reduction in the formation of micro-cracks, plastic deformation, and shattered particles on the worn surface of PA6/5GF composites. However, further addition of GFs significantly increased more grooves, plastic deformation, and broken particles of the worn surface.

3.10 | Morphological of wear surface analysis by SEM

Morphology of the worn surface of pure PA6 and PA6/GF(5, 10, and 20 mass%) based composites was analyzed using the SEM to investigate the topographical

attributes and wear mechanism. SEM images of worn surface of PA6 and PA6/GF(5, 10, and 20 mass%) reinforcement composite are shown in Figure 16. It indicates that the additive wear is crucial in interpreting the wear mechanism. When the PMC pins were sliding over the steel disc surface, the adhesion occurred in the surface of the steel material; as a result, the wear occurred more. Similar observations were made by other researchers where plastic deformation and peeling pits present on worn surface determined the adhesive wear.^{26,63} In Figure 16b, there was no micro-crack formation and broken particles which signifies the strong adhesion of the PA6 matrix to the GF reinforcement indicating the less adhesive wear. Additionally, the morphology of the PA6/5GF composite's surface was smoother as compared to the pristine PA6, indicating the formation of a stable and high strength protection layer on the top of PA6/5GF composite. In contrast, the worn out surfaces of pure PA6 and PA6/GF(10 and 20 mass%) reinforced composites exhibited many micro-crack, plastic deformation, and broken particles indicating the more adhesive wear, and are shown in the Figures 16a,c,d, respectively which attributed to the worst tribological properties. It has been determined that the particle emission and the wear were

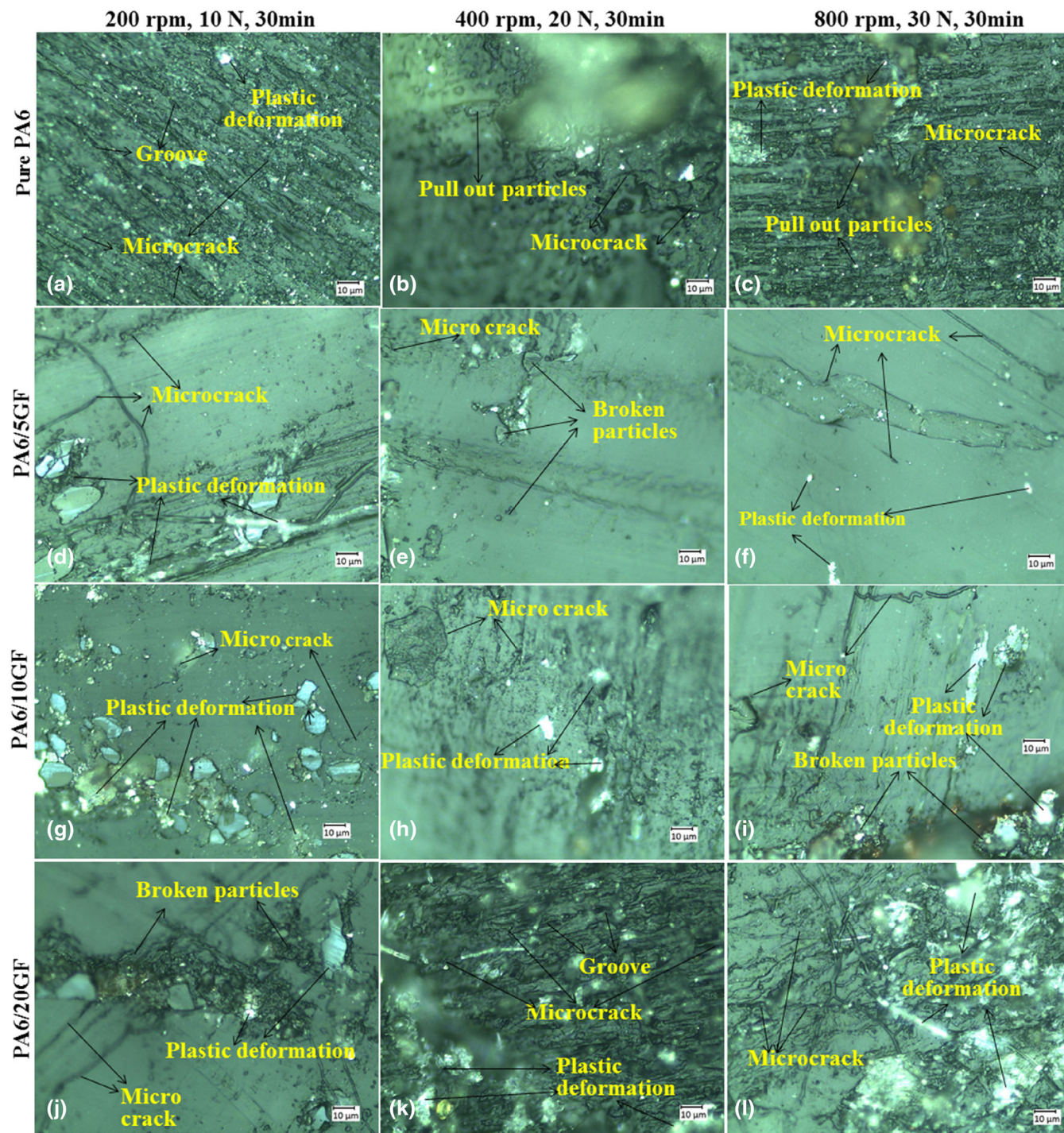


FIGURE 15 Optical microscopic image for tested worn pure PA6 and PA6/5GF, PA6/10GF, and PA6/20GF composites samples sliding against EN31 steel. GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

caused by plastic deformation and adhesive damage at the weak polymer matrix.²⁶

In the EDS analysis, main elemental compositions such as, the content of oxygen (O), carbon (C), nitrogen (N), magnesium (Mg), aluminium (Al), sulfur (S), chlorine (Cl), silicon (Si), and calcium (Ca) dispersed on the

surface of the pin samples are shown in Figure 17. From the EDS analysis, the Si content was found as 1.15, 1.37, and 2.23 mass% PA6/5GF, PA6/10GF, and PA6/20GF composites, respectively, and hence, it confirmed the increased amount of GF content present in the PA6/GF composites, respectively.

FIGURE 16 SEM images of worn surfaces for (a) PA6 and (b) PA6/5GF, (c) PA6/10GF, and (d) PA6/20GF composites samples. GF, glass fiber; SEM, scanning electron microscope. [Color figure can be viewed at wileyonlinelibrary.com]

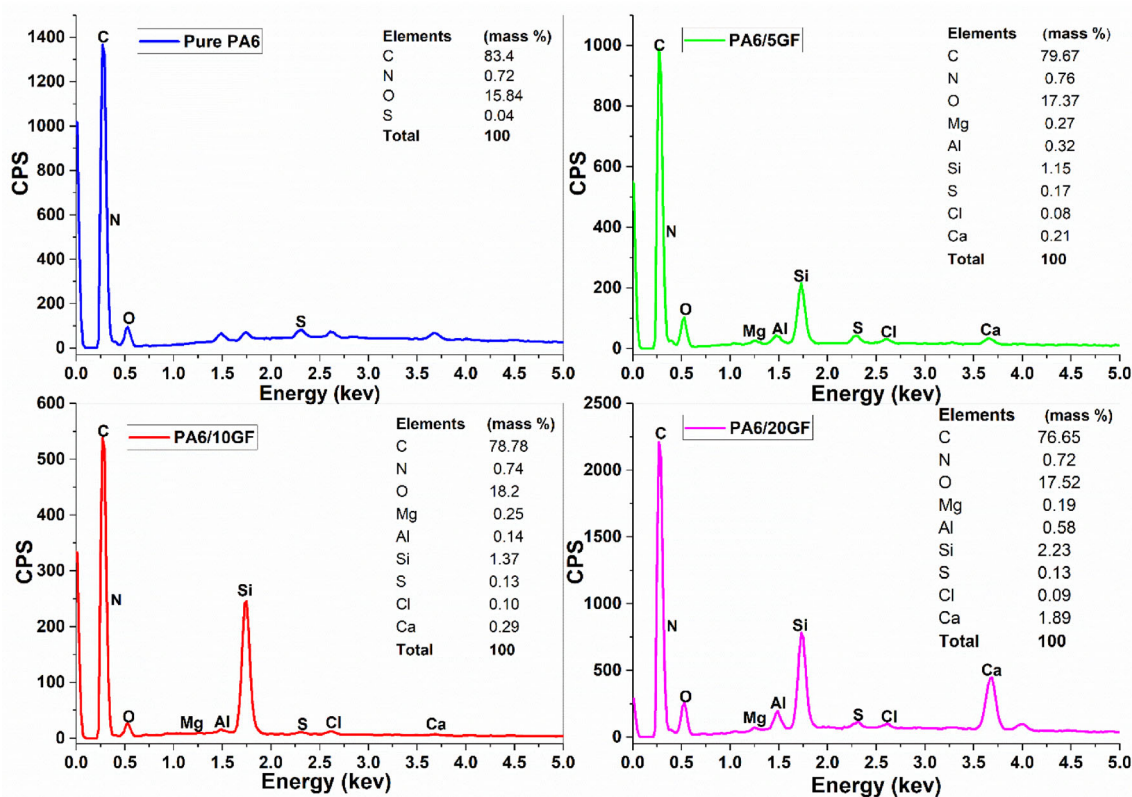
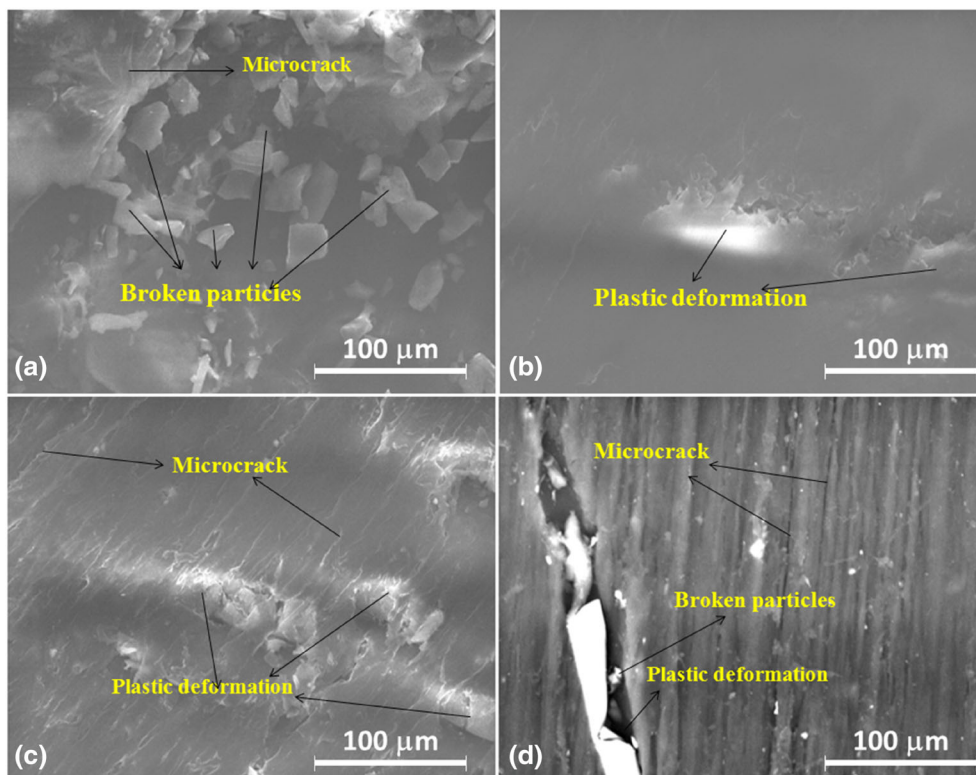


FIGURE 17 EDS of worm debris for PA6 and PA6/5GF, PA6/10GF, and PA6/20GF composites samples. GF, glass fiber; EDS, energy dispersive spectrum. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 8 Response table of S/N ratio for COF of PA6/GF(0, 5, and 10) composites.

The-smaller-the-better				
Level	Mass%	Time (min)	Load (N)	Speed (rpm)
1	4.809	7.055	7.997	7.729
2	11.392	8.566	8.464	8.251
3	8.513	9.094	8.254	8.735
Delta	6.583	2.039	0.468	1.006
Rank	1	2	4	3

Abbreviations: COF, coefficient of friction; GF, glass fiber.

TABLE 9 Response table of S/N ratio for COF of PA6/GF(0, 5, and 20) composites.

The-smaller-the-better				
Level	Mass%	Time (min)	Load (N)	Speed (rpm)
1	4.809	8.020	8.116	8.413
2	11.392	8.685	9.149	9.216
3	10.282	9.778	9.219	8.854
Delta	6.583	1.758	1.103	0.803
Rank	1	2	3	4

Abbreviations: COF, coefficient of friction; GF, glass fiber.

TABLE 10 Response table of mean COF for PA6/GF(0, 5, and 10) composites.

Level	Mass%	Time (min)	Load (N)	Speed (rpm)
1	0.5794	0.4729	0.4361	0.4448
2	0.2669	0.3858	0.3875	0.3996
3	0.3708	0.3583	0.3935	0.3727
Delta	0.3125	0.1146	0.0485	0.0720
Rank	1	2	4	3

Abbreviations: COF, coefficient of friction; GF, glass fiber.

TABLE 11 Response table of mean COF for PA6/GF(0, 5, and 20) composites.

Level	Mass%	Time (min)	Load (N)	Speed (rpm)
1	0.5794	0.4341	0.4311	0.4217
2	0.2669	0.3808	0.3645	0.3608
3	0.3039	0.3353	0.3547	0.3677
Delta	0.3125	0.0988	0.0764	0.0609
Rank	1	2	3	4

Abbreviations: COF, coefficient of friction; GF, glass fiber.

TABLE 12 Response table of S/N ratio of WR for PA6/GF(0, 5, and 10) composites.

The-smaller-the-better				
Level	Mass%	Time (min)	Load (N)	Speed (rpm)
1	96.89	107.58	108.22	106.21
2	115.87	108.18	106.84	111.64
3	111.35	108.35	109.06	106.26
Delta	18.98	0.78	2.22	5.44
Rank	1	4	3	2

Abbreviations: GF, glass fiber; WR, wear rate.

TABLE 13 Response table of S/N ratio of WR for PA6/GF(0, 5, and 20) composites.

The-smaller-the-better				
Level	Mass%	Time (min)	Load (N)	Speed (rpm)
1	96.89	105.41	110.32	106.17
2	115.87	110.28	106.80	109.48
3	111.25	108.32	106.89	108.36
Delta	18.98	4.87	3.51	3.30
Rank	1	2	3	4

Abbreviations: GF, glass fiber; WR, wear rate.

3.11 | Taguchi method

Based on Equation (2), the signal-to-noise (S/N) ratio of COF and WR has been determined for each experimental parameter. The effect of control factors on a given COF and WR was examined using the S/N response table. The parameters included reinforcement weight percentage, sliding time, applied normal load, and sliding speed. Tables 8–11 and Tables 12–15, respectively, illustrate the evaluation of process parameters using “S/N ratios” and “mean values” derived for various parameter levels for the COF and the WR. In term of the S/N ratio, the control variables are statistically significant. The “main effect

plots” for the “mean” and the “S/N ratio” of COF and WR are displayed in Figures 18 and 19, respectively. The impact of these parameters and levels of the responses is depicted by the “main effect plots” for S/N ratios. Since the highest S/N ratio consistently produces the best quality with the least amount of variance, the level with the highest value reveals would be the ideal level for each factor, according to MINITAB v16's consideration. The “response table” displays the “average reaction” attributed to each level of each factor in a design. The data of “rank” and “delta” can be used to evaluate the variables that have the biggest impact on the outcome. The following is an interpretation of the “delta” and “rank” values:

TABLE 14 Response table of mean WR for PA6/GF(0, 5, and 10) composites.

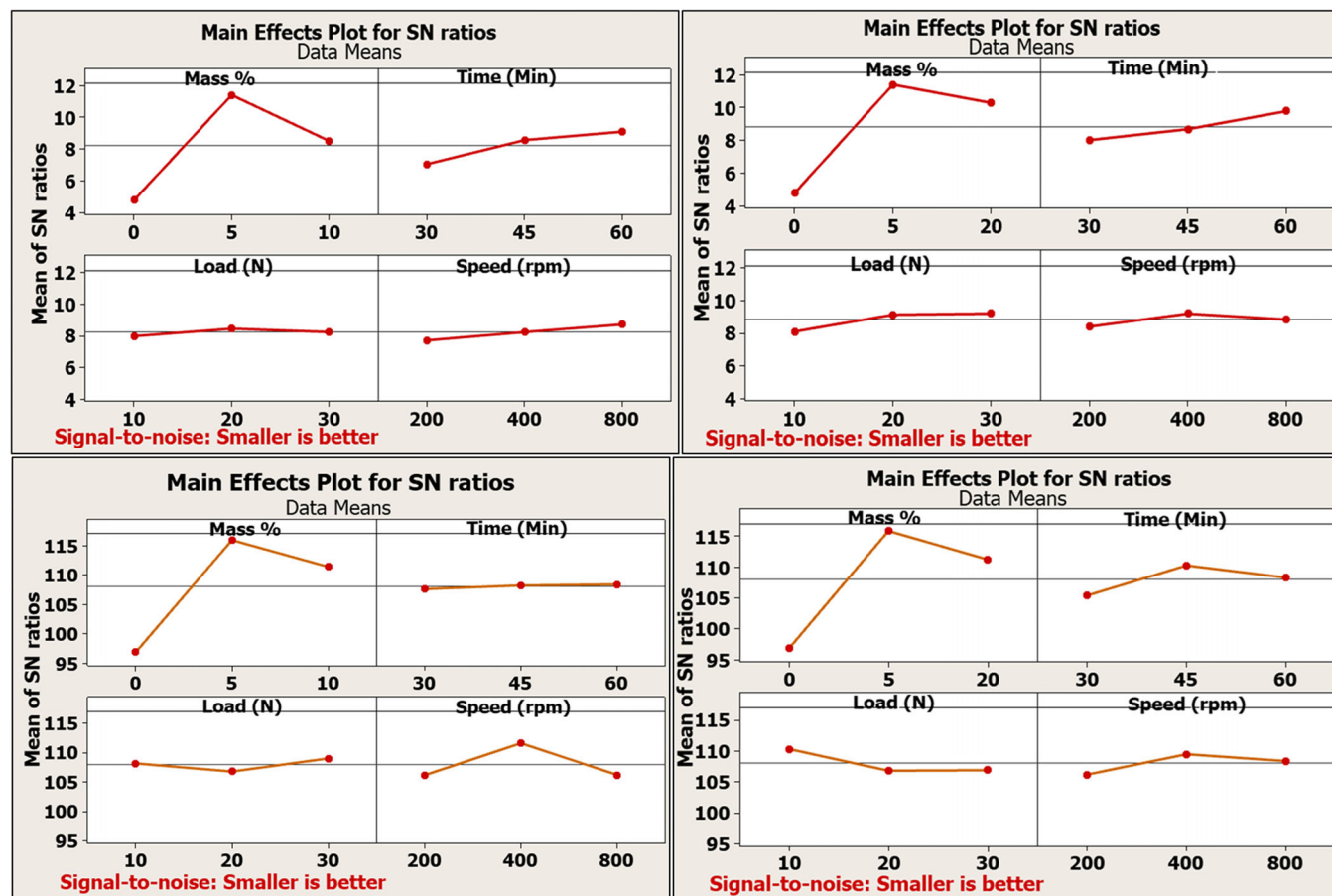
Level	Mass%	Time (min)	Load (N)	Speed (rpm)
1	0.000015	0.000007	0.000007	0.000008
2	0.000002	0.000005	0.000006	0.000004
3	0.000003	0.000007	0.000006	0.000007
Delta	0.000013	0.000002	0.000002	0.000003
Rank	1	3	4	2

Abbreviations: GF, glass fiber; WR, wear rate.

TABLE 15 Response table of mean WR for PA6/GF(0, 5, and 20) composites.

Level	Mass%	Time (min)	Load (N)	Speed (rpm)
1	0.000015	0.000008	0.000007	0.000008
2	0.000002	0.000005	0.000006	0.000005
3	0.000003	0.000007	0.000007	0.000006
Delta	0.000013	0.000003	0.000001	0.000003
Rank	1	2	4	3

Abbreviations: GF, glass fiber; WR, wear rate.

FIGURE 18 Main effect plot for COF mean and S/N ratio for PA6 and PA6/GF composites. COF, coefficient of friction; GF, glass fiber. [Color figure can be viewed at wileyonlinelibrary.com]

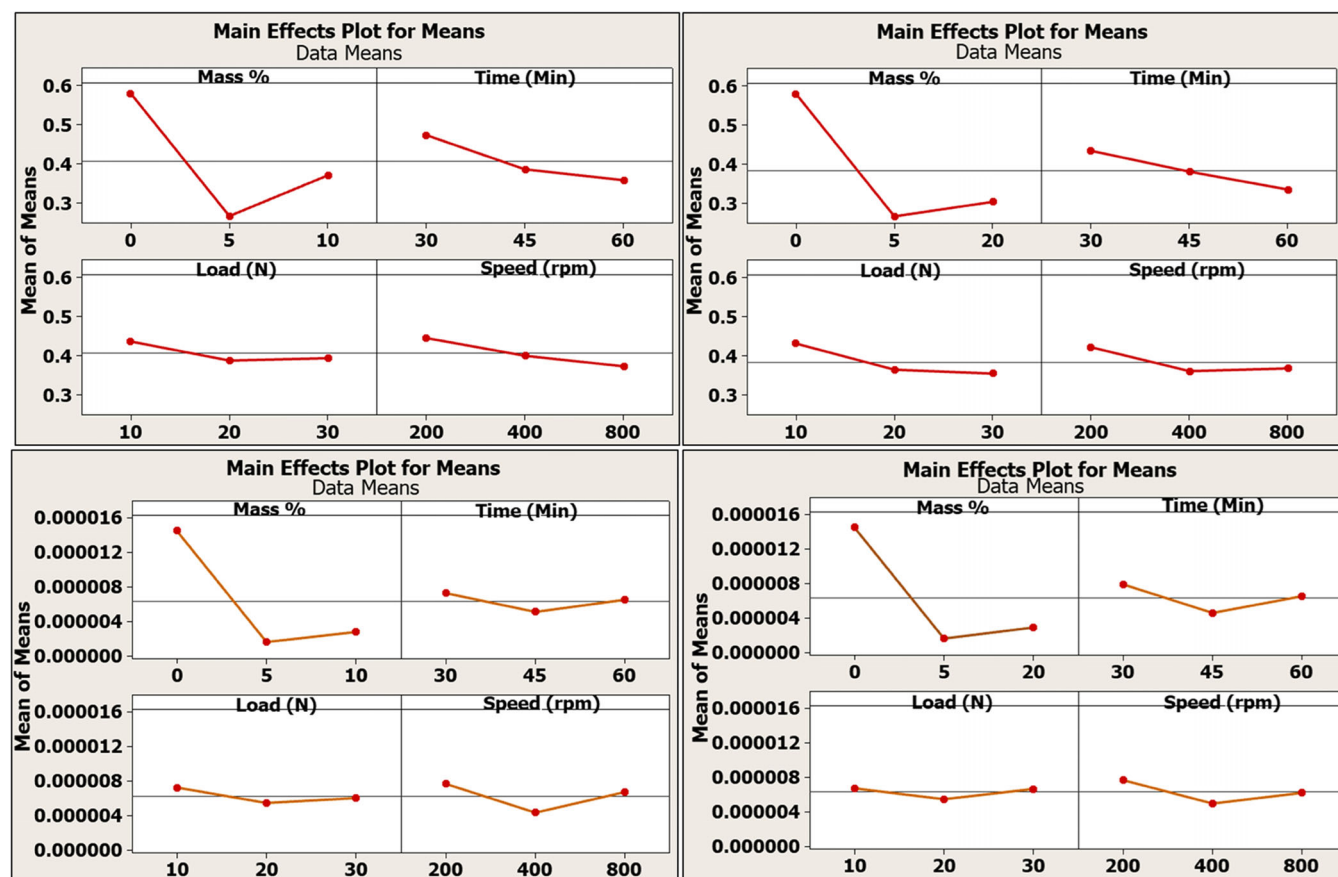


FIGURE 19 Main effect plots for mean WR and S/N ratio for PA6 and PA6/GF composites. GF, glass fiber; WR, wear rate. [Color figure can be viewed at wileyonlinelibrary.com]

“Delta”: the effect size is calculated by subtracting the extreme values from the average responses.

“Rank”: make a list of the influences, ordering them from most significant (based on the delta values) to least significant (on the characteristic response).

3.12 | Analysis of variance

The ANOVA on COF and WR for PA6/GF (0, 5, 10 and 20) composites is given in Table 16. Figure 20 shows a normal probability plot for COF and WR. This plot was made by testing the normality of the experimental results, and it shows how the predicted values for the design matrix compare to the actual values. In addition, residuals shown against expected plots are regularly distributed. Therefore, the established mathematical models can be applied efficiently for subsequent research, and

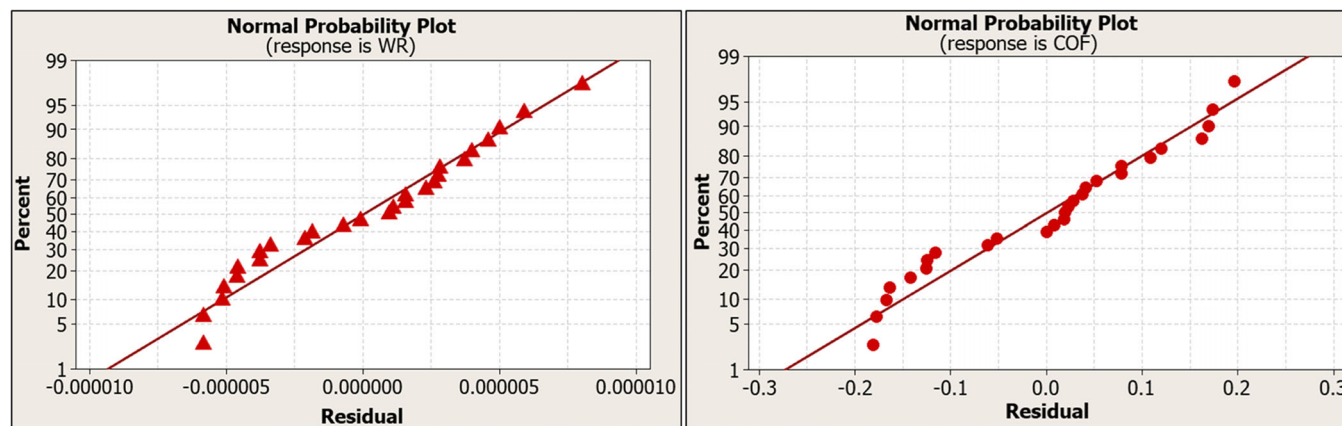
there is no reason to ensure independence is violated. It is essential for any ANOVA test that the normal probability plot should be closer to the mean line. As depicted in Figure 21, all experimental values have shown significantly closer to the mean line. The ANOVA table shows that the p -values, which are less than 0.05, indicate a statistically significant contribution to the performance metrics for both COF and WR.

The “interaction plot” results shown in Figure 21a,b depict the influence of reinforcement (in mass%), time (in min), normal load (in N), and velocity (in rpm). The normal load is a parameter that has a greatest influence on COF and WR. The “interaction plot” depicts the interaction effect of control factor on performance evaluation, demonstrating the minimum value of COF and WR at the interaction of 5 mass% GF on PA6 composites relative to PA6 composites and other composites. The “interaction plot” also shows that the COF and WR can be increased with increasing of the normal load for 20 N representing higher COF and WR, respectively. Similarly, the speed would be an influential factor for COF and WR, since 800 rpm showed greater COF and WR values.

TABLE 16 Analysis of variance (ANOVA) results of COF and WR for PA6/GF(0, 5, and 10) and PA6/GF(0, 5, and 20) composites.

Property and composite	ANOVA results			
	Source	Regression	Residual error	Total
COF for PA6/GF(0, 5, and 10) composites	DF	4	22	26
	SS	0.28429	0.36007	0.64436
	MS	0.07107	0.01637	
	<i>F</i>	4.34		
	<i>p</i>	0.010		
COF for PA6/GF(0, 5, and 20) composites	DF	4	22	26
	SS	0.28429	0.40725	0.68294
	MS	0.06892	0.01851	
	<i>F</i>	3.72		
	<i>p</i>	0.018		
WR for PA6/GF(0, 5, and 10) composites	DF	4	22	26
	SS	6.27981×10^{-10}	4.22797×10^{-10}	1.05078×10^{-9}
	MS	1.56995×10^{-10}	3.07294×10^{-11}	
	<i>F</i>	8.17		
	<i>p</i>	0.000		
WR for PA6/GF(0, 5, and 20) composites	DF	4	22	26
	SS	3.67778×10^{-10}	6.76048×10^{-10}	1.04383×10^{-9}
	MS	9.19445×10^{-11}	3.07294×10^{-11}	
	<i>F</i>	2.99		
	<i>p</i>	0.041		

Abbreviations: COF, coefficient of friction; GF, glass fiber; WR, wear rate.

**FIGURE 20** Normal probability plot for the COF and WR. COF, coefficient of friction; WR, wear rate. [Color figure can be viewed at wileyonlinelibrary.com]

4 | CONCLUSIONS

The objective of this article is to provide guidance to readers in choosing the suitable filler for a PA6 matrix that is capable of withstanding high loads, speeds, and temperatures. Due to its unique mix of stiffness and

durability, the material is ideal for many uses, including tie-rod gears, bearings, and as an important structural element in car bodies. Addition of GF(5, 10, and 20 mass%) into PA6 matrix enhanced the thermal, structural, tribological, and mechanical properties. It is concluded that the addition of 5 mass% GF into PA6

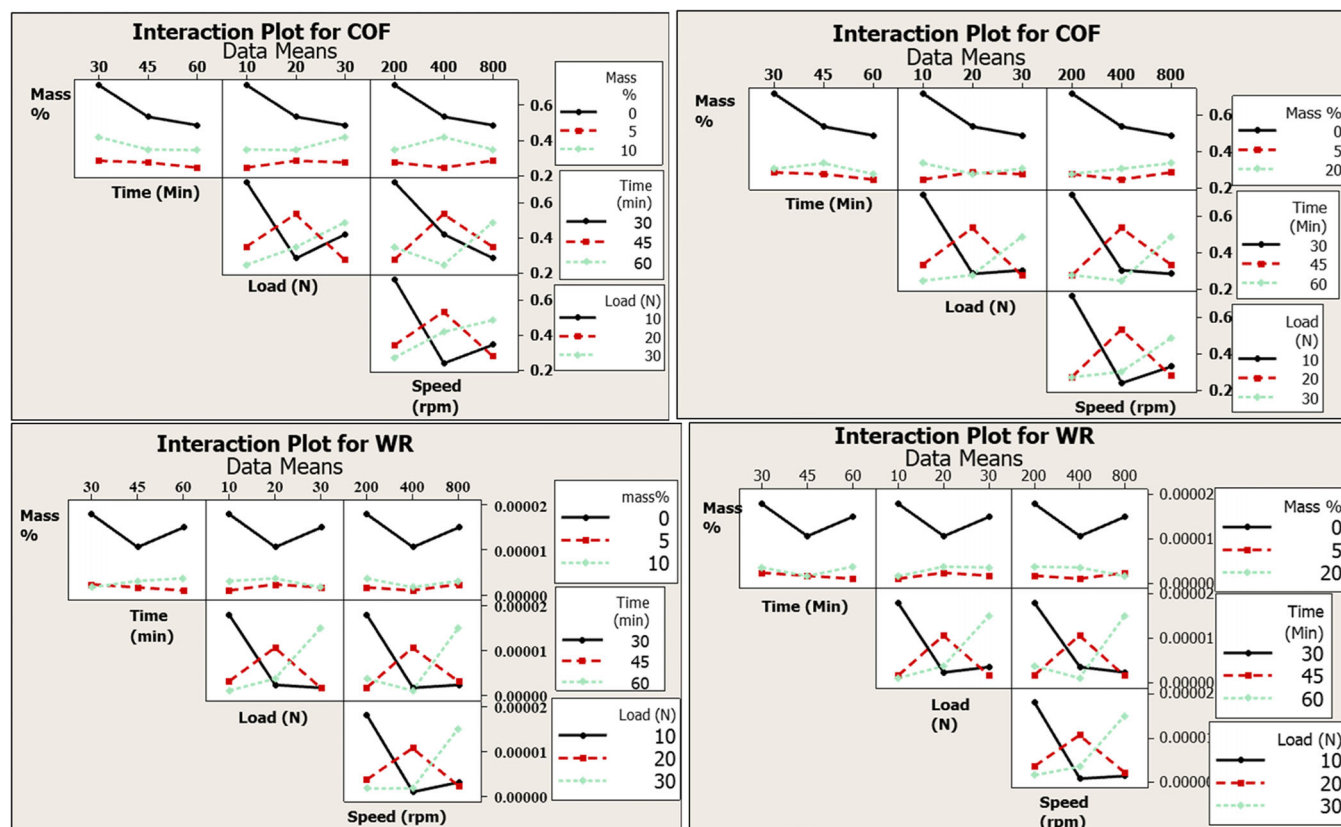


FIGURE 21 “Interaction effects” plot of COF and WR for PA6/GF(0, 5, 10, and 20) composites. COF, coefficient of friction; GF, glass fiber; WR, wear rate. [Color figure can be viewed at wileyonlinelibrary.com]

matrix results to 11.4% higher hardness ($RH_F = 45.20$) and 38.3% higher tensile strength ($UTS = 44.46$ MPa) compared to pure PA6 as well as significantly higher than those of other studies' composites.^{54,55} In addition, the activation energy (E_a) derived from thermal kinetic studies increased by 13% for PA6/5GF and 53% for PA6/10GF composites relative to the pure polymer matrix. The XRD and FTIR analyses indicate that the chopped GF fillers uniformly distributed in the PA6 matrix, as all main peaks of from both the phases were distinctly present in all the composites. The statistical study using ANOVA determined the impact of GF fillers on wear properties of PA6 matrix with 95% confidence interval. The ANOVA table results showed that the samples at specific loading condition having p -values of less than 0.05 indicate a statistically significant contribution to the performance metrics for both COF and WR characteristics. Especially, the normal load and speed showed as influential factor for the COF and WR values, which were greater at 20 N and 800 rpm. It can be concluded that the tribological properties of all developed composites especially, the average COF of PA6/5GF composite decreased by 17%–65% and its average WR significantly decreased by 68%–92%

with respect to pure PA6. Therefore, the present PA6/GF composites can be used in automobile, mechanical, and marine industries because of its light-weight and high wear resistance. Especially, it can be used to fabricate the tie-rod gears and bearings a crucial structural component in automotive bodywork.

The PA6/5 mass% GF composite was determined to be the most favorable composite among those materials analyzed in this study. The current investigation primarily focused on the tribological properties and other supportive properties of the materials. However, a detailed analysis of various functional properties, such as surface tension, impact, and nanostructures, will be conducted in our future study.

AUTHOR CONTRIBUTIONS

Kamlendra Vikram: Data curation (lead); formal analysis (equal); investigation (equal); writing – original draft (equal). **Sumit Pramanik:** Conceptualization (lead); data curation (supporting); formal analysis (lead); investigation (equal); methodology (lead); project administration (lead); supervision (lead); writing – original draft (equal); writing – review and editing (lead). **Shubrajit Bhaumik:** Conceptualization (supporting); methodology

(equal); project administration (equal); supervision (supporting); writing – review and editing (supporting).

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CONFLICT OF INTEREST STATEMENT

Authors declare no competing interests that are relevant to the content of this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Sumit Pramanik  <https://orcid.org/0000-0003-0455-1582>

REFERENCES

- [1] K. Friedrich, *Adv. Ind. Eng. Polym. Res.* **2018**, 1, 3.
- [2] M. E. Çetin, A. C. Tatar, O. Demir, G. Önal, A. Avcı, *Mech. Mater.* **2021**, 155, 103780.
- [3] M. E. Çetin, *Polym. Compos.* **2021**, 42, 4353.
- [4] A. Pogačnik, A. Kupec, M. Kalin, *Wear* **2017**, 378, 17.
- [5] K. K. Kar, S. Pramanik, Hydroxyapatite poly(etheretherketone) nanocomposites and method of manufacturing same. U.S. Patent No. 8652373 B2, **2014**.
- [6] M. Palabiyik, S. Bahadur, *Wear* **2000**, 246, 149.
- [7] O. Starkova, Z. Zhang, H. Zhang, H.-W. Park, *Mater. Sci. Eng., A* **2008**, 498, 242.
- [8] R. Fathi, A. Ma, B. Saleh, Q. Xu, J. Jiang, *Mater. Today Commun.* **2020**, 24, 101169.
- [9] S. Kumar, K. Panneerselvam, *Procedia Technol.* **2016**, 25, 1129.
- [10] H. Unal, A. Mimaroglu, *Wear* **2012**, 289, 132.
- [11] R. Kumar, S. K. Mishra, S. M. Hasnain, S. Pandey, A. F. Deifalla, S. Jayapalan, *Mater. Res. Express* **2023**, 10, 105306.
- [12] M. J. Clifford, T. Wan, *Polymer* **2010**, 51, 535.
- [13] S.-H. Wu, F.-Y. Wang, C.-C. M. Ma, W.-C. Chang, C.-T. Kuo, H.-C. Kuan, W.-J. Chen, *Mater. Lett.* **2001**, 49, 327.
- [14] S. Zhang, J. Zhang, L. Tang, J. Huang, Y. Fang, P. Ji, C. Wang, H. Wang, *Polymer* **2019**, 11, 978.
- [15] A. Grozdanov, G. Bogoeva-Gaceva, *J. Thermoplast. Compos. Mater.* **2010**, 23, 99.
- [16] K. Q. Han, Z. J. Liu, M. H. Yu, *Macromol. Mater. Eng.* **2005**, 290, 688.
- [17] H. C. Cartledge, C. A. Baillie, *J. Mater. Sci.* **1999**, 34, 5099.
- [18] A. Greco, A. Maffezzoli, E. Calò, C. Massaro, R. Terzi, *J. Therm. Anal. Calorim.* **2012**, 109, 1493.
- [19] D.-X. Li, X. Deng, J. Wang, J. Yang, X. Li, *Wear* **2010**, 269, 262.
- [20] S. S. Kim, M. W. Shin, H. Jang, *Wear* **2012**, 274, 34.
- [21] M. M. Y. Zaghloul, K. Steel, M. Veidt, M. T. Heitzmann, *Tribol. Int.* **2024**, 191, 109059.
- [22] I. N. Uysal, M. A. Tasdelen, *Macromol. Res.* **2024**, 32, 663.
- [23] H. Khakbaz, A. P. Basnayake, A. K. A. Harikumar, M. Firouzi, D. Martin, M. Heitzmann, *Polym. Degrad. Stab.* **2024**, 227, 110870.
- [24] M. M. Y. Zaghloul, K. Steel, M. Veidt, D. Martin, M. Firouzi, M. T. Heitzmann, *Tribol. Lett.* **2023**, 71, 102.
- [25] W. Nie, J. Li, X. Sheng, *Mater. Sci. Pol.* **2010**, 28, 67.
- [26] G. Tang, W. Huang, D. Chang, W. Nie, W. Mi, W. Yan, *Polym. Plast. Technol. Eng.* **2011**, 50, 1537.
- [27] K. Vikram, S. Bhaumik, S. Pramanik, *J. Therm. Anal. Calorim.* **2023**, 148, 3341.
- [28] A. Balbosh, M. Kern, *J. Prosthet. Dent.* **2006**, 95, 218.
- [29] R. Mosharrar, P. Ranjbarian, *J. Adv. Prosthodont* **2013**, 5, 126.
- [30] Q. Ji, M. Zhang, M. Rong, B. Wetzel, K. Friedrich, *Tribol. Lett.* **2005**, 20, 115.
- [31] K. Vikram, S. Pramanik, S. Bhaumik, *Iran. Polym. J.* **2024**, 33, 511.
- [32] M. E. Çetin, Y. Baştosun, A. C. Tatar, M. H. Cetin, O. Demir, G. Önal, A. Avcı, *Polym. Compos.* **2022**, 43, 624.
- [33] C. Gologlu, N. Sakarya, *J. Mater. Process. Technol.* **2008**, 206, 7.
- [34] V. Gupta, V. Kothari, *Manufactured Fibre Technology*, Springer Science & Business Media, Bristol, UK **1997**.
- [35] J. Workman Jr, L. Weyer, *Practical Guide to Interpretive Near-Infrared Spectroscopy*, CRC press, Boca Rotan, USA **2007**.
- [36] K. Nakamoto, *Infrared and Raman Spectra of Inorganic and Coordination Compounds, Part B: Applications in Coordination, Organometallic, and Bioinorganic Chemistry*, John Wiley & Sons, New Jersey, USA **2009**.
- [37] K. Nishikida, J. Coates, *Handbook of Plastics Analysis*, CRC Press, Boca Raton, USA **2003**, p. 198.
- [38] J. Coates, *Encyclopedia of Analytical Chemistry*; John Wiley & Sons Ltd, Chichester, UK **2000**, p. 10815.
- [39] N. Vasanthan, D. Salem, *J. Polym. Sci., Part B: Polym. Phys.* **2001**, 39, 536.
- [40] C. Isbasar, J. Hacaloglu, *J. Anal. Appl. Pyrolysis* **2012**, 98, 221.
- [41] J.-C. Ho, K.-H. Wei, *Macromolecules* **2000**, 33, 5181.
- [42] T. M. Majka, M. Cokot, K. Pielichowski, *J. Therm. Anal. Calorim.* **2018**, 131, 405.
- [43] F. Najafizadeh, M. A. S. Sadjadi, S. J. Fateami, M. K. Mobarakeh, R. M. Afshar, *Hacettepe J. Biol. Biochem.* **2016**, 44, 317.
- [44] Q. Meng, Y. Gu, L. Luo, S. Wang, M. Li, Z. Zhang, *J. Appl. Polym. Sci.* **2017**, 134, 44832.
- [45] P. Wilson, S. Vijayan, K. Prabhakaran, *J. Ind. Eng. Chem.* **2019**, 80, 401.
- [46] R. Singh, S. M. Desai, G. Pathak, *J. Appl. Polym. Sci.* **2003**, 87, 2146.
- [47] X. Fu, X. Dong, G. Yang, S. J. H. Bai, *Heliyon* **2022**, 8, 8.
- [48] A. B. Kulkarni, S. N. Mathad, R. P. Bakale, *Ovidius Univ. Ann. Chem.* **2019**, 30, 60.
- [49] C. Li, K. Suzuki, *J. Therm. Anal. Calorim.* **2009**, 98, 261.
- [50] V. Kumar, M. Kumar, G. Pugazhenthii, *Int. J. Nano Biomater.* **2014**, 5, 27.
- [51] X. Zhang, *Eng. Sci.* **2020**, 14, 1.
- [52] N. Abacha, M. Kubouchi, T. Sakai, *eXPRESS Polym. Lett.* **2009**, 3, 245.

- [53] Y. F. Chen, T. Zhang, M. Tang, D. Xie, Q. Long, C. Y. Li, *E-Polymers* **2017**, 17, 23.
- [54] H. Essabir, D. Rodrigue, R. Bouhfid, A. el Kacem Qaiss, *Polym. Compos.* **2018**, 39, 14.
- [55] D. M. Nuruzzaman, A. Iqbal, N. M. Ismail, M. Y. Ali, A. Iqbal, N. Hossain, *AIP Conf. Proc.* **2023**, 2643, 050062.
- [56] A. Meram, M. E. Cetin, *J. Mater. Eng. Perform.* **2020**, 29, 7408.
- [57] M. Watanabe, H. Yamaguchi, *Wear* **1986**, 110, 379.
- [58] L. Chang, Z. Zhang, L. Ye, K. Friedrich, *Tribol. Int.* **2007**, 40, 1170.
- [59] M. Sharma, J. Bijwe, P. Mitschang, *Wear* **2011**, 271, 2261.
- [60] N. G. Karsli, S. Demirkol, T. Yilmaz, *Composites, Part B* **2016**, 88, 253.
- [61] S. Zhou, Q. Zhang, C. Wu, J. Huang, *Mater. Des.* **2013**, 44, 493.
- [62] Y.-L. You, D.-X. Li, G.-J. Si, R.-Y. Lv, X. Deng, *J. Thermoplast. Compos. Mater.* **2016**, 29, 494.
- [63] M. Wan, S.-E. Li, H. Yuan, W.-H. J. C. P. A. A. S. Zhang, *Composites, Part A* **2019**, 117, 34.

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